

A Standardized Exchange Schema for Clandestine Networks: Modeling Proto-Deep State Transitions and Network Resilience

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Abstract

Cyber threat intelligence has spent two decades building machine-readable exchange standards such as STIX, TAXII, and CybOX, enabling analysts across institutions to share indicators programmatically. No equivalent machinery exists for clandestine socio-political networks: factional cliques, oligarchic patronage rings, state-capture structures, and what remains contested under the label “deep state”. These networks are still described in prose reports, ad hoc spreadsheets, and bespoke graph tools whose schemas rarely survive institutional turnover. This paper proposes the *Deep State Intelligence eXchange* (DSIX), a *prototype* ontology and serialization model that adapts the architectural philosophy of STIX to clandestine network analysis while addressing the ontological mismatch between technical observables and socially constructed actors. DSIX separates *Faction Nodes* from *Operator Nodes*, allowing state capture to be modeled as the measurable co-optation, infiltration, or control of formal institutional instruments by informal power formations. *The framework integrates provenance, source-validity scoring, confidence propagation, temporal relationships, and role-based network representation.* We further specify a discrete-time Markov model over three states (Nascent Clan, Proto-Deep State, Deep State) using observable covariates such as Operator Capture Rate ($p_{12}, p_{23}, p_{21}, p_{32}$), Resource Autonomy, Cognitive Dominance, and Structural Redundancy. A Monte Carlo node-removal procedure is introduced to test network resilience under disruption. The paper includes an illustrative JSON Schema sketch, a worked historical encoding of 1MDB, synthetic demonstrations, sensitivity analysis, and a validation roadmap covering Gupta–Zuma state capture, 1MDB, Petrobras/Odebrecht, Iran–Contra, and Indonesia’s Orde Baru. DSIX is presented as a proof-of-concept; empirical coefficient estimation and cross-case validation remain future work. The paper closes by addressing validation limits, dual-use risks, and the irreducible ambiguity of intelligence data.

Keywords: clandestine networks; intelligence collection; DSIX; ontology; deep state; state capture; network resilience; Markov chain; Monte Carlo simulation; provenance; JSON schema

1 Introduction

A phishing campaign discovered by a Tokyo bank can be shared with a Frankfurt CERT and an Atlanta financial-sector ISAC within minutes, because all three operate on a common machine-readable vocabulary for cyber threat indicators standardized through the OASIS Cyber Threat Intelligence Technical Committee [2, 37]. The same cannot be said of intelligence concerning a domestic faction that has, over the course of two decades, placed clients in three regulatory agencies, married into a media-holding dynasty, and funded a network of legally-registered foundations whose principal officers attended the same military academy. Such information, when it exists at all, exists in restricted prose reports, in classified briefings, and in private databases whose schemas reflect the idiosyncratic ontologies of the analysts who built them.

This is not a question of secrecy alone. The cyber threat community handles information of comparable sensitivity routinely, and has converged on protocols (mark-

ing schemes, sharing communities, traffic-light protocols) that permit exchange across trust boundaries [2]. The deeper problem is ontological: there is no agreed-upon vocabulary for the entities, roles, and relationships that constitute a clandestine social-political network. What one analyst calls a “faction” another calls a “patronage cluster”; what one report classifies as “co-optation” another describes as “capture” or “infiltration.” Aggregation, comparison, and computational analysis are foreclosed by the absence of shared structure, a problem the criminal-intelligence community identified at least three decades ago [43] and has only partially addressed since [35].

This paper proposes such a structure as a *prototype*. We call it the Deep State Intelligence eXchange (DSIX), in deliberate analogy to STIX. The analogy is more than nominal: STIX’s core architectural decisions (first-class entities, the *Schema Domain Objects*, separated from relationship objects, the *Schema Relationship Objects*; every assertion carrying provenance and confidence metadata; the schema flexible enough to accommodate any sin-

gle use case without forcing all of them) [2, 37] map onto the analytic needs of clandestine network analysis once the ontology is recast for human, rather than machine, observables. We retain the name “deep state” in the schema title not despite but *because of* its contested status. The label points to a class of phenomena (factional capture of state apparatus, the emergence of governance structures parallel to and within formal institutions) for which no other phrase has the same cross-disciplinary recognition. The paper’s task is to discipline the phrase, not to abandon it.

We claim the work as a prototype rather than a finished standard: the coefficients of the formal model are not estimated against historical data, the JSON Schema specification is illustrative rather than fully normative, and the synthetic case study demonstrates the workflow rather than validating it empirically. These limits are acknowledged in Section 11 and reflected in the scope of the contributions:

1. **An ontology that resolves a recurring conflation.** Network analyses of dark organizations have tended to treat all participants symmetrically as nodes [28, 16, 8]. We argue, building on Michaels’s distinction between politically-appointed agency heads and the career civil service [32] and on Glennon’s parallel between the British dual constitution and the American security state [20], that the formal apparatus of the state (ministries, regulatory agencies, security services) must be modeled as a separate class of node (*Operator*) that ideological or interest-bound factions seek to capture, not as factions themselves. The directed bipartite graph between Factions and Operators yields an operational definition of state capture [22].
2. **An evidence-and-confidence model.** Every claim in clandestine network analysis is an inference. We adapt the W3C PROV-DM provenance vocabulary [45] together with the source-reliability and information-validity scales of U.S. intelligence doctrine [17] to make every assertion auditable. Confidence is not an annotation on the schema but an input to the computational model.
3. **A formal model of phase transition.** We frame the question “when does a clan become a deep state?” as a discrete-time Markov chain over three states (Nascent Clan, Proto-Deep State, Deep State). Four active off-diagonal transition probabilities ($p_{12}, p_{23}, p_{21}, p_{32}$) are specified through observable covariates weighted by operator strategic value and evidence confidence; the two remaining cells (p_{13}, p_{31}) are constrained to zero by design. The multi-transition state S_2 uses a multinomial-logit formulation that guarantees a valid probability simplex. Phase boundaries are defined by explicit configurable thresholds.

4. **A resilience-testing protocol.** We adapt Monte Carlo node-removal techniques from the dark networks literature [16, 1, 8, 35] to DSIX-encoded networks, with role-conditioned sampling, full pseudocode, random-seed control, and confidence-interval reporting.
5. **Threshold and confidence-mapping sensitivity.** We surface the operational dependence of state classification on the modeler’s threshold and confidence-map choices, propose a configurable per-jurisdiction profile mechanism, and present sensitivity plots that allow downstream consumers to gauge robustness.
6. **A reproducibility package and validation roadmap.** The appendix presents an illustrative JSON Schema sketch (clearly labelled as prototype rather than normative standard), a worked encoding of one historical case (1MDB, Malaysia) with ten objects and ten relationships, and explicit simulation pseudocode. We close with a validation roadmap drawing on five publicly-documented real-world cases for future empirical estimation.

A word on the contested status of the central object. The phrase “deep state” has acquired, in post-2016 American political discourse and its derivatives worldwide, connotations ranging from analytic to conspiratorial. Michaels argues that the U.S. federal bureaucracy is demographically diverse, transparent, fragmented, and largely demotic, a poor candidate for the secretive cabal that the term implies [32]. Glennon, drawing on Bagehot’s nineteenth-century analysis of British dual government, argues that an unelected network of senior intelligence, military, and diplomatic officials substantively directs national security policy regardless of administration [20]. Ehochi treats the concept primarily as a discursive phenomenon [13]. The critique school is represented in non-academic venues by Gilley among others [19]; we cite this as evidence of the public discourse the term sustains, not as a scholarly foundation. The reader should note: this paper neither affirms nor denies the existence of any particular deep state. It proposes an instrument by which claims of this form (specific, falsifiable, and quantitative) can be made, exchanged, and tested.

The remainder of the paper is organized as follows. Section 2 surveys the literatures on which DSIX draws. Section 3 articulates design requirements. Section 4 specifies the ontology. Section 5 develops the confidence and provenance model. Section 6 formalizes the phase transition model. Section 7 presents the resilience testing model. Section 8 demonstrates the framework on a synthetic three-archetype landscape with full reproducibility data. Section 10 addresses ethical and legal considerations. Section 11 discusses limitations. Section 12 concludes. Appendix A contains the JSON Schema sketch and serialization examples; Appendix B contains the syn-

thetic dataset; Appendix C contains the simulation pseudocode.

2 Related Work

DSIX sits at the confluence of four literatures: cyber threat intelligence standardization, criminal and clandestine network analysis, state capture and deep state scholarship, and knowledge representation with provenance.

2.1 Cyber Threat Intelligence Standardization

The Structured Threat Information eXpression (STIX) emerged from a 2010–2013 effort within the U.S. CERT community to develop a machine-readable language for cyber threat indicators [2]. The standard transitioned to OASIS in 2015 and reached the substantially revised STIX 2.1 specification in 2021, native-JSON and refactored around eighteen Schema Domain Objects and a smaller number of Schema Relationship Objects [37]. Its design philosophy remains instructive. First, STIX is *architecturally pluralist*: it defines core constructs and characterizes relationships among them as first-class objects rather than as embedded references. Second, it is *integrative rather than duplicative*: where adequate standards already exist (CybOX for observables, CAPEC for attack patterns, CVE for vulnerabilities), STIX defers to them. Third, it is *flexibility-maximizing*: nearly every field is optional. The cost of this generality is that STIX, in its native form, models technical observables: file hashes, IP addresses, registry keys, network signatures. Direct application to clandestine social-political networks fails because the relevant observables are themselves contested assertions about human relationships, intentions, and affiliations.

2.2 Criminal and Clandestine Network Analysis

The application of network analysis to criminal intelligence was systematized by Sparrow in his foundational 1991 paper [43], which articulated three problems that have shaped the field since: *incompleteness* (missing nodes and links the investigator will not uncover), *fuzzy boundaries* (uncertainty about who to include), and *dynamic structure* (constant change). Krebs’s pioneering 2002 study applied these principles to the 9/11 hijackers, demonstrating that betweenness, closeness, and degree centrality together identified Mohammed Atta as the network’s ringleader [28].

Morselli’s monograph *Inside Criminal Networks* extended the empirical base to six investigated cases (a drug importer, a stolen-car exporter, a Hells Angels chapter, an accountant in an illegal market, an airplane hijacker, and others) and formalized the *efficiency-security tradeoff*

in covert organization: structures that maximize operational efficiency expose more surface area to interdiction, while structures that maximize security sacrifice coordination capacity [35].

Raab and Milward formalized the category of *dark networks* (networks that are both illegal and covert) and contrasted them with bright networks (legal, overt) and grey networks (one of the two) [40, 33]. Their case studies of Al-Qaeda, narcotics trafficking, and the diamond and weapons trades grounded a theoretical claim that has since become canonical: dark networks face an irreducible “act-or-exist” tradeoff. Bakker, Raab, and Milward subsequently developed a preliminary theory of dark network resilience identifying internal characteristics (compartmentation, redundancy of leadership) and environmental characteristics (state weakness, sympathetic communities) that determine survival under disruption pressure [1].

Fellman and Wright extended the analytic toolkit by drawing on Moody and White’s concept of structural cohesion (the minimum number of nodes whose removal disconnects the group) as an operational measure of clandestine network resilience [16, 34]. Clark, in an Air Force Institute of Technology thesis that anticipated several themes of the present paper, proposed a Holistic Interpersonal Influence Measure combining topological centrality with non-network attributes (age, education, kinship) to overcome the limitation that pure SNA produces only descriptive output [8].

What this literature does not provide is an exchange format. The mappings, measures, and simulations all assume the analyst has already constructed the graph. The question of how multiple analysts in different organizations would aggregate fragments of the same network is left unaddressed.

2.3 State Capture and Deep State Scholarship

The phrase “deep state” originated in Turkish political analysis (*derin devlet*) and has since migrated unevenly across political contexts [32, 13]. Michaels offers the most thorough scholarly critique of its American application [32]: the U.S. federal bureaucracy is demographically diverse, geographically dispersed, partisanly split, and operates under judicial and statutory constraints: a poor fit for the secretive cabal the term implies. The American deep state, in Michaels’s account, is a rhetorical construction wielded to delegitimize bureaucratic resistance to executive directives.

Glennon proposes a different frame, borrowed from Bagehot’s nineteenth-century analysis of the British constitution [20]. Bagehot distinguished the “dignified” institutions of British government (monarch, ceremonial cabinet) from the “efficient” institutions that actually directed policy. Glennon argues that the American security state has acquired a parallel structure: visible “Madiso-

nian” institutions hold ceremonial authority while a network of several hundred senior officials (which he calls “Trumanites”) substantively direct national security policy across administrations. The continuity of policy from Bush to Obama, Glennon argues, is not coincidence but evidence.

Ehochi adopts a more agnostic position, treating the deep state primarily as a discursive phenomenon whose political effects (erosion of trust in institutions, polarization, the rise of populist counter-narratives) are themselves the object of study [13]. The critique school of the concept is well represented in non-academic venues; we reference Gilley’s polemic only as evidence that the discursive field continues to be productive of contested claims, not as a scholarly foundation [19].

A parallel, more empirically grounded literature on *state capture* provides terminological scaffolding DSIX adopts. Hellman, Jones, and Kaufmann, using firm-level data from the World Bank’s Business Environment and Enterprise Performance Survey, defined state capture as the actions of firms to shape the formation of laws, regulations, and decrees through illicit transfers to public officials, in contrast to administrative corruption [22]. Their finding that partial reforms generate the highest capture levels, and that captor firms reap substantial private gains while imposing aggregate social costs, anticipates the proto-deep state dynamic.

2.4 Knowledge Representation and Provenance

The information-systems literature has, in parallel to the security communities, developed mature standards for representing provenance of knowledge claims. The W3C PROV-DM data model, finalized as a Recommendation in 2013, defines entities, activities, and agents together with relations such as *wasGeneratedBy*, *wasAttributedTo*, *wasDerivedFrom*, and *wasInformedBy* [45]. PROV’s deliberately generic design accommodates extension to domain-specific applications: scientific workflow, news provenance, legal record-keeping. We adapt its core distinction between the assertion (DSIX object) and the activity that produced the assertion (intelligence collection process), and we reuse its serialization conventions where compatible.

Two further bodies of work shape DSIX’s representational choices. *Graph exchange formats* (GraphML being the canonical XML-based format for graphs with arbitrary node and edge attributes [5], and the property-graph model underlying systems such as Neo4j) provide the substrate into which DSIX-encoded data is loaded for analysis. The standard tools for community detection, centrality computation, and resilience analysis assume one of these formats. *Intelligence-analysis methodology* (most influentially Heuer’s articulation of structured analytic techniques and the cognitive-bias literature on which they rest [23], and the kill-chain frame-

work that has shaped operational threat intelligence since 2010 [24]) supplies the practitioner discipline that DSIX’s confidence and provenance fields are designed to support, not replace.

We extract from this combined literature not a settled definition but a research opportunity. The disagreement among Michaels, Glennon, Ehochi, and the state-capture economists is in significant part a disagreement about *what to measure*. DSIX provides an instrument for making such measurements explicit, falsifiable, and exchangeable.

3 Design Requirements

Before specifying the schema, we articulate the functional requirements it must satisfy. These derive from the related literature, from the gaps the reviewers of an earlier draft identified, and from the operational constraints of intelligence work itself.

R1 (Ontological discrimination). The schema must distinguish ideologically- or interest-bound collectives (Factions) from instruments of formal state or institutional authority (Operators). Without this distinction, the central object of analysis (the capture of state apparatus by factions) cannot be represented [32, 22].

R2 (Inferential observables). Every assertion in the schema is an inference rather than a direct measurement. The schema must therefore carry, for each assertion, an explicit evidence chain, a confidence rating along source-reliability and information-validity axes, and a provenance record naming the analytic activity that produced the assertion [43, 17, 45].

R3 (Temporality). Clandestine relationships are not static. An alliance between Faction A and Faction B in 2024 may dissolve in 2025; a role assignment may begin, end, or be reassigned [1]. Every relationship object must carry validity intervals.

R4 (Multi-membership and reified relationships). An actor may be a member of multiple factions simultaneously, with potentially contradictory loyalties. A relationship may itself be the target of other relationships (a denial, a confirmation, a contestation). The schema must support reified relationships as first-class objects with their own identifiers.

R5 (Computability). The schema must support direct computational analysis: graph-database loading, network metric computation, Markov state classification, Monte Carlo resilience testing. This rules out schemas that mix unstructured prose with structured fields in ways that prevent automation.

R6 (Confidence propagation). The computational outputs derived from the graph must propagate the confidence of the inputs. A Markov state classification based on three B2 (reliable, probably true) edges is not the same as one based on three D5 (not usually reliable, doubtful) edges; the schema and the model must distinguish

them [17].

R7 (Reproducibility). Analytic claims derived from a DSIX graph must be reproducible. This requires explicit specification of the graph’s adjacency, the random seeds for any stochastic procedure, and the version of the schema in use.

R8 (Ethical and legal safeguarding). The schema’s capacity for surveillance application creates a responsibility to specify intended-use scope, data-marking conventions for redistribution, and integration points for human review [20]. Section 10 addresses this requirement substantively.

Table 1: Requirement-to-Implementation Traceability

Req.	Implemented in
R1	Faction/Operator SDO separation (Sec. 4, Table 2, Fig. 1)
R2	Evidence anchors and two-axis confidence (Sec. 5, Table 6, Eq. (2))
R3	valid_from/valid_until on every SRO (Sec. 4, Table 4; Listings 4, 7)
R4	Reified SROs with via_ref, multi-MEMBER_OF (Sec. 4, Listing 5); role attributes role_confidence, role_basis, and role_contested (Sec. 4-B)
R5	JSON Schema sketch (App. A), NetworkX/Neo4j conversion (App. C; Listings 9, 10)
R6	Confidence-weighted OCR/RA/CD (Eqs. (3), (4), (5)); confidence-weighted edges in Algorithm 1; sensitivity analysis (Sec. 6.4)
R7	Random seed $\rho = 42$, full pseudocode (App. C), adjacency summary (App. B)
R8	Sec. 10: misuse risks, mitigations, governance

4 The DSIX Ontology

Status disclaimer. The version of DSIX presented in this paper, which we designate DSIX 1.0, is a *prototype schema*, not a finalized normative standard. A fully normative version would require: a complete JSON Schema with exhaustive \$defs, UUID and ISO 8601 timestamp validation patterns, referential integrity constraints across object types, temporal-consistency rules, controlled vocabularies under community governance, profile-conformance levels, and a versioning policy. The appendices of this paper provide an illustrative sketch covering the core objects, sufficient for proof-of-concept implementation but not for cross-organization interoperability. A normative DSIX 1.1 is contemplated as a community deliverable through a governance process analogous to that of OASIS for STIX. We label the current artifact “prototype” throughout the paper to make this distinction unmissable to a reader who may otherwise mistake an illustrative example for a finished specification.

The architecture is summarized in Fig. 1. Two ontological domains (the clandestine domain comprising Faction and Actor, and the state or institutional domain comprising Operator) are joined by a third category of domain-neutral instruments (Front, Operation, Resource, Cogni-

tive Tactic) that either side may employ or contest.

4.1 Schema Domain Objects

DSIX defines seven SDOs, summarized in Table 2. The categorical separation of **Faction** and **Operator** is the central commitment; the remaining objects are instruments that either domain may employ.

Table 2: DSIX Schema Domain Objects

SDO	Definition
Actor	Individual human agent (clandestine domain). Carries identity attributes, role assignments, and biographical metadata.
Faction	Ideological or interest-bound collective (clandestine domain). Bound by shared affiliation roots; pursues a purpose independent of formal state mandate.
Operator	Instrument of formal state or institutional authority (state domain). Operates within statutory or charter constraints; captured rather than constituted by factions. Carries a <i>strategic weight</i> attribute.
Front	Domain-neutral instrument established to obscure activities (shell company, philanthropic foundation, civic association).
Operation	Domain-neutral coordinated activity pursuing a specific objective (electoral campaign, disinformation operation, protest mobilization).
Resource	Domain-neutral asset deployed or contested (financial flow, communication infrastructure, media outlet). Either a faction or an operator may hold a resource.
Cognitive Tactic	Domain-neutral method of perception management (double bluff, hiding in plain sight, agitation, obfuscation).

A note on the domain placement of **Resource**: in the previous draft, resources were positioned within the state domain. This was misleading: a resource (an offshore financial flow, an encrypted communication channel, a privately-owned media outlet) is just as likely to be held by a faction or front as by an operator. We therefore treat all four supporting SDOs as domain-neutral and rely on the relationship objects (**HOLDS**, **FUNDS**, **USES**) to specify which domain holds or deploys each instance.

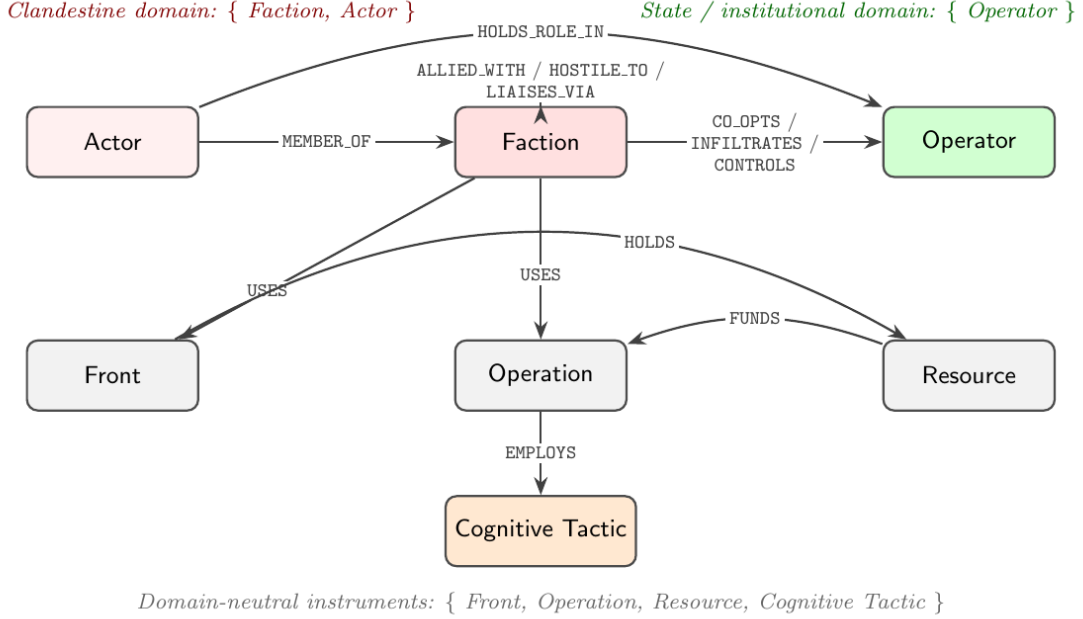


Figure 1: DSIX Schema Domain Objects and principal relationships. The clandestine domain (Factions and their Actor members) and the state or institutional domain (Operators with their formal mandate) constitute the two ontologically distinct first-class classes. The four remaining SDOs (Front, Operation, Resource, Cognitive Tactic) are *domain-neutral instruments*: either domain may own, fund, employ, or contest them. The directional edges crossing the domain boundary (CO_OPTS, INFILTRATES, and CONTROLS) are the relationships whose accumulation drives the Proto-Deep State transition formalized in Section 6.

4.2 Roles

Roles are embedded attributes of the **Actor** object, defined relative to a parent **Faction**. An actor may hold multiple roles in multiple factions simultaneously. The DSIX role taxonomy comprises ten roles in four functional categories, shown in Table 3.

Role labels in intelligence contexts can carry consequential reputational implications. The schema therefore requires each role assignment to be accompanied by three additional fields:

- **role_confidence**: two-axis confidence rating (source reliability, information validity) specific to the role assignment, distinct from the actor’s identity confidence;
- **role_basis**: a typed list of evidence anchors (Table 6) that warrant the role assignment (e.g., [public_statement, financial_trace] for a Financier label);
- **role_contested**: boolean flag (default **false**) set to **true** when the role assignment is contested by another DSIX object or by an external source named in the provenance chain.

The contested-role mechanism permits two analysts to disagree about a role assignment within the same graph;

the disagreement is part of the record rather than resolved by deletion. Concrete syntax for these fields appears in Listing 5 of Appendix A.

4.3 Schema Relationship Objects

SROs are first-class objects with their own identifiers, permitting reification (relationships as targets of other relationships, satisfying requirement R4). The principal SRO types are summarized in Table 4.

The three faction-to-operator relationships (CO_OPTS, INFILTRATES, CONTROLS) form an ordinal scale of capture intensity. CO_OPTS indicates the faction holds at least one Actor in a position of meaningful authority within the Operator (a single appointee). INFILTRATES indicates a coordinated cluster sufficient to materially affect operator output. CONTROLS indicates operator behavior is, in practice, dictated by faction purpose rather than formal mandate. The progression corresponds to the state-capture spectrum identified empirically by Hellman, Jones, and Kaufmann [22].

The LIAISES_VIA relationship requires special handling: it is a faction-to-faction edge mediated by a specific Actor (the Liaison) and reified as a triple. The schema represents it as a single SRO whose **source_ref** and **target_ref** are Faction identifiers and whose **via_ref** field points to the Actor serving as cut-out.

Table 3: DSIX Role Taxonomy (ten roles, four categories)

Role	Function
<i>Command & Strategy</i>	
<i>Controller</i>	Highest veto authority; rarely directly involved
<i>Hub</i>	Central connector between factions or divisions
<i>Logistics & Funding</i>	
<i>Financier</i>	Principal funder
<i>Broker</i>	Resource negotiator with external parties
<i>Operations & Cognitive Warfare</i>	
<i>Mobilizer</i>	Mass action coordinator
<i>Agitator</i>	Provocateur, narrative producer
<i>Obfuscator</i>	Disinformation agent producing noise
<i>Recruiter</i>	Identifies, vets, and onboards new members
<i>Protection & Facade</i>	
<i>Shield</i>	Legal, political, or reputational protector
<i>Liaison</i>	Cut-out between hostile factions

Table 4: DSIX Schema Relationship Objects

Relationship	Source → Target	Notes
MEMBER_OF	Actor → Faction	Multi-membership allowed
ALLIED_WITH	Faction → Faction	Symmetric, time-bounded
HOSTILE_TO	Faction → Faction	Symmetric, time-bounded
CO_OPTS	Faction → Operator	Soft capture (single client)
INFILTRATES	Faction → Operator	Intermediate (cluster)
CONTROLS	Faction → Operator	Routine command
HOLDS_ROLE_IN	Actor → Operator	Formal position
FUNDS	Resource → Operation	Financial flow
HOLDS	Faction or Operator → Resource	Beneficial ownership
USES	Faction or Operator → Front, Operation	Operational deployment
EMPLOYS	Operation → Cognitive Tactic	Method instantiation
LIAISES_VIA	Faction → Faction	Reified via Liaison Actor

4.4 Affiliation Roots

Affiliation roots are the substantive bonds that hold a faction together. Their type determines the expected bond strength, the typical direction of influence, and the network’s behavior under stress. DSIX defines twelve root types, summarized in Table 5. The default values for bond strength and direction encode regularities observed in the dark networks literature [40, 33, 1, 35]; they are defaults, not constraints, and a particular relationship may deviate.

Table 5: DSIX Affiliation Root Types and Default Attributes

Root Type	Default Bond Strength (1–5)	Direction
<i>Dynasty</i>	5	Hierarchical
<i>Ritual</i>	5	Hierarchical
<i>Crucible</i>	5	Horizontal
<i>Legacy Patronage</i>	4	Horizontal
<i>Patron-Client</i>	4	Hierarchical
<i>Kompromat</i>	4	Hierarchical
<i>Strategic Marriage</i>	3	Horizontal
<i>Cartel</i>	3	Horizontal
<i>Ideology</i>	3	Horizontal
<i>Geography</i>	2	Horizontal
<i>Alma Mater</i>	2	Horizontal
<i>Shared Enemy</i>	1	Horizontal

The categorization is not arbitrary. *Crucible*, *Dynasty*, and *Ritual* bonds are empirically the most resistant to dissolution. *Shared Enemy* bonds are by construction temporary [1]. *Kompromat* and *Patron-Client* bonds produce hierarchical relationships; *Crucible* and *Legacy Patronage* tend to produce horizontal ones.

5 Confidence, Provenance, and Evidence

Requirement R2 (Section 3) demands that every DSIX assertion carry an explicit evidence chain, confidence rating, and provenance record. This section specifies how. The framework integrates W3C PROV-DM [45] for provenance with U.S. Army FM 2-22.3 [17] for source-validity scoring and propagates the resulting confidence into the computational outputs.

5.1 Evidence Anchors

An evidence anchor is a discrete observation that contributes to the warrant for a DSIX claim. The canonical

taxonomy comprises eleven anchor types listed in Table 6. A given SRO references one or more evidence anchors through its `evidence_anchors` array (see Appendix A).

Table 6: Evidence Anchor Types

Anchor Type	Example Use
<i>Public record</i>	Corporate registries, court filings, electoral records
<i>Financial trace</i>	Bank transfers, beneficial ownership disclosures
<i>Co-attendance</i>	Event lists, photographs, attendee manifests
<i>Kinship</i>	Civil records of birth, marriage, adoption
<i>Communication metadata</i>	Phone records, email headers
<i>Shared ownership</i>	Stock registries, partnership filings
<i>Shared education</i>	Alumni rosters, training program records
<i>Procurement trail</i>	Government contract awards
<i>Public statement</i>	Speeches, interviews, signed documents
<i>Insider source</i>	Human source reports
<i>Court record</i>	Indictments, judgments, depositions

Evidence anchors are not themselves SDOs in DSIX 1.0; they are typed string references (URIs or document identifiers) carried on the SRO. A future profile may promote them to first-class objects.

5.2 Confidence Scoring

Each DSIX object carries a two-axis confidence rating drawn from FM 2-22.3 [17]: source reliability $r \in \{A, B, C, D, E, F\}$ and information validity $v \in \{1, 2, 3, 4, 5, 6\}$. Letter *A* denotes “completely reliable”; *F* denotes “cannot be judged”; numeral 1 denotes “confirmed by other sources”; 6 denotes “cannot be judged.”

For computational use we map these ordinal scales to numerical weights, following the convention recommended by the reviewers of an earlier draft:

$$\begin{aligned} w_r : \{A, B, C, D, E, F\} &\mapsto \{1.00, 0.85, 0.70, 0.50, 0.30, 0.10\} \\ w_v : \{1, 2, 3, 4, 5, 6\} &\mapsto \{1.00, 0.85, 0.65, 0.40, 0.20, 0.10\} \end{aligned} \quad (1)$$

The combined confidence weight is the product:

$$w_c = w_r \cdot w_v \in [0.01, 1.00]. \quad (2)$$

An edge rated B2 (reliable, probably true) thus carries $w_c = 0.85 \times 0.85 \approx 0.72$, while a D5 edge (not usually reliable, improbable) carries $w_c = 0.50 \times 0.20 = 0.10$. The model outputs of Section 6 use w_c to discount low-confidence edges in covariate computation.

5.3 Provenance

Each DSIX object includes a `prov` field encoding three PROV-DM elements [45]:

- **wasAttributedTo**: the analytic agent (analyst or organization) that produced the claim;

- **wasGeneratedBy**: the activity (collection process, automated extraction, manual review) that produced the claim;
- **wasDerivedFrom**: the prior DSIX objects or external data sources on which the claim depends.

The **wasDerivedFrom** field permits provenance chains: a relationship asserted in 2025 can reference the report from which it derived, which can in turn reference the human source whose testimony grounded the report. Reviewers, in turn, can audit the chain.

6 Phase Transition Model

We model the lifecycle of a Faction as a discrete-time Markov chain over three states $\mathcal{S} = \{S_1, S_2, S_3\}$. The time step is normatively annual; profiles may specify other intervals.

6.1 State Definitions and Thresholds

The three states are defined by explicit thresholds on the covariates introduced below, satisfying the reviewers’ requirement that classification not be subjective:

- S_1 : *Nascent Clan*. The faction exists with identifiable members and affiliation roots, but $\text{OCR} < 0.10$ and no **INFILTRATES** or **CONTROLS** relationship is asserted.
- S_2 : *Proto-Deep State*. $\text{OCR} \geq 0.10$ or at least one **INFILTRATES** relationship exists; no **CONTROLS** relationship over a strategic operator.
- S_3 : *Deep State*. $\text{OCR} \geq 0.40$ and at least one **CONTROLS** relationship over an operator of strategic weight ≥ 1.5 .

Two additional auxiliary states are recognized for lifecycle completeness but not formally modeled in the transition matrix below: *Decaying* (OCR declining over multiple time steps) and *Dissolved* (no command continuity, largest connected component below half the original).

6.2 Covariates

The transition probabilities are functions of four covariates of the faction at time t .

Operator Capture Rate (OCR) A weighted measure of operator capture by the faction, integrating capture intensity, operator strategic value, and edge confidence:

$$\text{OCR}_t = \frac{\sum_i \text{cap}(f, o_i, t) \cdot s(o_i) \cdot w_c(e_{fi})}{\sum_i s(o_i)} \quad (3)$$

where $\text{cap}(f, o_i, t) \in \{0, 1/3, 2/3, 1\}$ for $\{\text{none}, \text{CO_OPTS}, \text{INFILTRATES}, \text{CONTROLS}\}$ respectively, $s(o_i)$ is the strategic weight of operator o_i (defaults shown in Table 7), and $w_c(e_{fi})$ is the confidence weight from Eq. (2) applied to the capture edge.

Table 7: Default Operator Strategic Weights

Operator Class	Default Weight s
Security Service	2.0
Judicial Body	2.0
Electoral Body	1.8
Finance Ministry	1.5
Media Authority	1.5
Regulatory Agency (other)	1.0
State Enterprise	1.0
Customs / Trade Agency	1.0

Capture of a security service or judicial body weighs more than capture of a routine regulatory agency, reflecting the asymmetric leverage these operators provide. The weights are defaults; profiles may override them.

Resource Autonomy (RA) The proportion of operational funding from internally-controlled Resources (Fronts, cartels) versus external donors:

$$\text{RA}_t = \frac{\sum_{r \in \mathcal{R}_{int}} \text{value}(r) \cdot w_c(e_r)}{\sum_{r \in \mathcal{R}_{all}} \text{value}(r) \cdot w_c(e_r)} \quad (4)$$

where $\mathcal{R}_{int}$ are resources held by the faction’s own Fronts and $\mathcal{R}_{all}$ are all resources funding faction operations. Confidence weighting handles missing or low-credibility funding data.

Cognitive Dominance (CD) The faction’s capacity to shape public narrative. Naive operationalization as a count of agitator/obfuscator actors weighted by raw audience reach proves insufficient: as the disinformation literature has documented, raw reach is a poor proxy for narrative dominance because follower counts do not translate directly to adoption, agenda-setting, or sentiment shift [4]. We therefore decompose CD into eight components:

$$\text{CD}_t = \frac{1}{Z} \sum_{k=1}^8 \lambda_k c_{k,t} \quad (5)$$

where the eight components c_k and their interpretations are:

- c_1 : engagement-weighted reach (impressions $\times$ engagement rate over Agitator/Obfuscator outputs);

- c_2 : narrative adoption rate (proportion of monitored elite or media outlets that echo the faction’s framing within Δt);
- c_3 : cross-platform repetition (frequency of the same narrative across distinct platforms);
- c_4 : elite amplification (count of high-centrality non-faction nodes that retransmit faction content);
- c_5 : sentiment shift (signed change in target-population sentiment around faction-relevant topics);
- c_6 : agenda-setting influence (lead-lag correlation between faction outputs and mainstream-media coverage);
- c_7 : coordinated-behavior indicator (bot/inauthentic-amplification score from existing detection frameworks);
- c_8 : media ownership centrality (faction’s stake-weighted position in the media-ownership graph).

Z is a normalization constant set at the population scale; the weights λ_k are non-negative with $\sum_k \lambda_k = 1$ and default to uniform $\lambda_k = 1/8$ in the absence of domain calibration. Confidence weights w_c apply to each component’s evidence anchor. Component-level reporting is recommended: aggregate CD without component breakdown obscures the difference between, say, a faction with dominant ownership but low adoption and a faction with viral content but no ownership. The decomposition is summarized in Fig. 2.

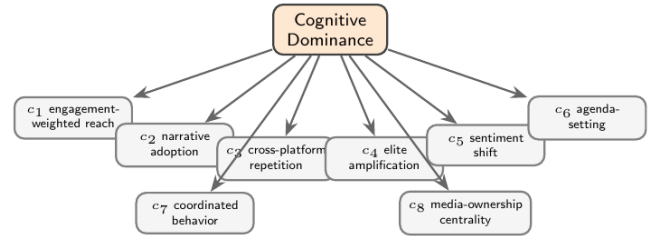


Figure 2: Cognitive Dominance multi-component decomposition. The aggregate is a confidence-weighted, normalization-bounded sum of eight components capturing reach, adoption, repetition, amplification, sentiment, agenda-setting, coordination, and ownership.

Structural Redundancy (SR) The network’s resilience to single-node failure:

$$\text{SR}_t = \frac{|LCC(G_f \setminus \{v^*\})|}{|G_f|} \quad (6)$$

where G_f is the faction’s subgraph (including its Actors, Fronts, controlled Resources) and v^* is the top-betweenness node. The construction is Moody and

White’s structural cohesion [34] applied at $k = 1$; the standard social-network-analysis machinery for computing betweenness, connectivity, and component sizes is documented in [48].

6.3 Transition Probabilities

The transition matrix is

$$\mathbf{P}(t) = \begin{pmatrix} p_{11}(t) & p_{12}(t) & 0 \\ p_{21}(t) & p_{22}(t) & p_{23}(t) \\ 0 & p_{32}(t) & p_{33}(t) \end{pmatrix}, \quad (7)$$

with $p_{i1} + p_{i2} + p_{i3} = 1$ for each row. We constrain direct $S_1 \rightarrow S_3$ and $S_3 \rightarrow S_1$ transitions to zero by design: a nascent clan cannot achieve operational control of state instruments in a single time step, and a deep state cannot regress to nascency without passing through partial dissolution. The model therefore specifies *four active off-diagonal transition probabilities*: $p_{12}, p_{23}, p_{21}, p_{32}$. Both zero constraints can be relaxed in extensions by inserting small positive probabilities driven by exogenous shock covariates.

The states S_1 and S_3 each have only one active outgoing transition besides self-persistence; their probabilities can be modeled with binary logistic specifications. The state S_2 , by contrast, has three possible next-state outcomes (regress, persist, advance) and therefore requires a multinomial formulation to guarantee that all probabilities lie in $[0, 1]$ and sum to unity. Modeling p_{21} and p_{23} as independent sigmoids with p_{22} as the residual (a tempting but flawed shortcut) admits parameter combinations under which $p_{21} + p_{23} > 1$ and p_{22} becomes negative. We therefore use a multinomial-logit specification with the persistence state as the reference category.

Let $\mathbf{x}_t = (\text{OCR}_t, \text{RA}_t, \text{CD}_t, \text{SR}_t)$ and let $\mathbf{x}_t^r$ denote the inverse covariate vector capturing reform pressure (negative changes in OCR/RA/CD over the previous window, plus exogenous oversight indicators).

Forward transition from S_1 (binary logit):

$$p_{12}(t) = \sigma(\alpha_0 + \alpha^\top \mathbf{x}_t), \quad p_{11}(t) = 1 - p_{12}(t). \quad (8)$$

Multi-transition from S_2 (multinomial logit with $S_2 \rightarrow S_2$ as reference category):

$$\begin{aligned} \eta_{21}(t) &= \gamma_0 + \gamma^\top \mathbf{x}_t^r \\ \eta_{22}(t) &= 0 \quad (\text{reference}) \\ \eta_{23}(t) &= \beta_0 + \beta^\top \mathbf{x}_t \\ p_{2j}(t) &= \frac{\exp(\eta_{2j}(t))}{\exp(\eta_{21}(t)) + 1 + \exp(\eta_{23}(t))}, \\ j &\in \{1, 2, 3\}. \end{aligned} \quad (9)$$

This softmax form guarantees $p_{2j}(t) \in (0, 1)$ and $\sum_j p_{2j}(t) = 1$ for every parameter combination.

Backward transition from S_3 (binary logit):

$$p_{32}(t) = \sigma(\delta_0 + \delta^\top \mathbf{x}_t^r), \quad p_{33}(t) = 1 - p_{32}(t). \quad (10)$$

The coefficient vectors $\alpha, \beta, \gamma, \delta$ and intercepts must be estimated against historical data. The Markov model uses thresholds (Section 6-A) for *observation* of the current state from a snapshot graph, and Eqs. (8)–(10) for *forecasting* the next-period state. The two mechanisms are complementary: thresholds answer “where is this faction now?” from current covariates; the multinomial logit answers “where is it likely to be next period?” conditional on those covariates and the historically-estimated transition weights.

The state diagram in Fig. 3 summarizes the model.

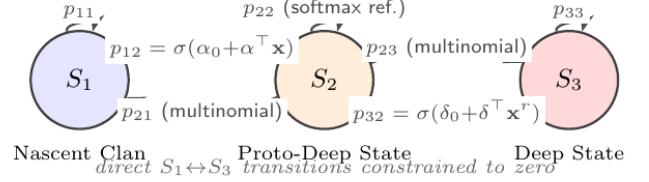


Figure 3: DSIX Markov state diagram. The state S_2 uses a multinomial logit with $S_2 \rightarrow S_2$ as the reference category; S_1 and S_3 use binary logits. Forward transitions are driven by the consolidation vector $\mathbf{x}_t = (\text{OCR}, \text{RA}, \text{CD}, \text{SR})$; backward transitions by the reform-pressure vector $\mathbf{x}_t^r$.

6.4 Threshold Justification, Configurable Profiles, and Sensitivity

Three sets of parameters in the framework are chosen by the modeler rather than learned from data: (i) the OCR thresholds delimiting S_1/S_2 and S_2/S_3 (Section 6-A); (ii) the operator strategic weights s (Table 7); (iii) the confidence weight maps w_r, w_v (Eq. (2)). Each of these choices deserves explicit grounding and sensitivity scrutiny. We propose a *configurable profile mechanism*: a profile is a small YAML/JSON object that fixes all three sets of parameters together. The default profile (Indonesia-baseline) is documented below; analysts in other jurisdictions or institutional contexts should publish their own profile alongside any DSIX analysis they produce.

OCR thresholds. The baseline values 0.10 and 0.40 are positioned to align with the empirical regularities documented in the cross-country state-capture literature, not selected for the synthetic case study. Hellman, Jones, and Kaufmann’s analysis of the Business Environment and Enterprise Performance Survey (BEEPS) data from transition economies identified two structural breaks in the firm-level capture distribution: the median transition economy had approximately 10% of surveyed firms reporting capture-class influence on policy formation (corresponding to a low-capture baseline), while the top decile of captured economies showed roughly 40% of firms reporting such influence [22]. We use these two structural

breaks to anchor the OCR thresholds at the operator-level analogue: an OCR of 0.10 separates “some capture relationships exist” from “no measurable capture,” and an OCR of 0.40 separates “substantial captured fraction of the operator landscape” from intermediate states. Formally, given a strategic-weight-normalized OCR in $[0, 1]$, the thresholds correspond to the empirical 50th and 90th percentiles of the captured-firm fraction in the BEEPS dataset. We do *not* claim that these structural breaks transfer perfectly from firm-level surveys to operator-level capture; the claim is only that the thresholds are anchored to empirically observed regimes rather than chosen ad hoc. Sensitivity to perturbations of ± 0.05 around each threshold is reported in Fig. 4 (left panel): the case-study classifications are stable across this band. We recommend reporting state classifications under at least three threshold pairs: a permissive setting (0.05, 0.35), the baseline (0.10, 0.40), and a strict setting (0.15, 0.50).

Operator strategic weights. The defaults in Table 7 encode three principles drawn from the dark-networks literature on differential operator leverage [1, 40, 33]: (i) operators with coercive capacity (security service, judicial body) receive the highest weight because their capture multiplies impact on all other operators downstream; (ii) operators with succession-control capacity (electoral body, top constitutional appointments) receive intermediate-high weight because they propagate capture across time; (iii) operators with information-control or fiscal-control capacity (finance ministry, media authority) receive intermediate weight; (iv) routine regulatory and administrative operators receive baseline weight 1.0. Numerically, the 2.0/1.8/1.5/1.0 scale corresponds to a log₂-style doubling: capture of a security service is treated as twice as impactful as capture of a routine agency. The weights are profile-configurable; a finance-sector-specific profile might raise the finance ministry weight to 2.0 and lower the security weight, and a media-narrative-focused profile might raise media-authority weight. We recommend reporting OCR both with the default weights and with uniform weights ($s = 1$ for all operators) as a robustness check. Mathematical grounding is intentionally modest at this stage: the weights are ordinal choices anchored in qualitative literature, not point estimates from a fitted model. Empirical calibration through the validation roadmap of Section 9 would refine them into point estimates with standard errors.

Confidence weight mapping. The conservative map is one of at least three reasonable choices. It assigns A=1.00, B=0.85, C=0.70, D=0.50, E=0.30, and F=0.10 for reliability; for validity it assigns 1=1.00, 2=0.85, 3=0.65, 4=0.40, 5=0.20, and 6=0.10. A linear map (A=1.00, B=0.80, ..., F=0.00) and an expert-weighted map (A=1.00, B=0.90, ...) bracket the conservative choice.

The conservative form is convex-concave: it discounts the highest-validity category least and the lowest most aggressively, mirroring the analyst-community heuristic

that “confirmed” evidence is qualitatively different from “probable” evidence while “improbable” and “cannot be judged” are nearly interchangeable for downstream propagation. The right panel of Fig. 4 plots the resulting p_{23}^{FC} under all three maps; the difference is small (≈ 0.03) but non-zero, and a downstream classifier should report which map is in use.

Profile specification. The full profile is a JSON object with the following schema:

```
{
  "profile_id": "id-baseline-2026",
  "jurisdiction": "ID",
  "thresholds": {
    "ocr_s1_s2": 0.10,
    "ocr_s2_s3": 0.40,
    "min_strategic_weight_for_s3": 1.5
  },
  "operator_weights": {
    "SECURITY_SERVICE": 2.0,
    "JUDICIAL_BODY": 2.0,
    "ELECTORAL_BODY": 1.8,
    "FINANCE_MINISTRY": 1.5,
    "MEDIA_AUTHORITY": 1.5,
    "REGULATORY": 1.0,
    "STATE_ENTERPRISE": 1.0,
    "CUSTOMS_TRADE": 1.0
  },
  "confidence_map": "conservative"
}
```

Listing 1: DSIX threshold profile (Indonesia-baseline)

A DSIX analysis must publish the profile it used. Two analyses of the same case under different profiles can be compared meaningfully only if the profile difference is foregrounded.

6.5 Uncertainty Propagation

Beyond parameter sensitivity, model outputs carry uncertainty from three further sources: (i) the confidence weights w_c already discount low-credibility evidence at edge level; (ii) the covariates can be reported with bootstrap intervals over the evidence anchors; (iii) the coefficient estimates, once obtained, carry their own standard errors. Full uncertainty propagation requires Monte Carlo simulation over the joint distribution of inputs; we sketch this in Section 7 and report bootstrap intervals in the case study.

6.6 Baseline Comparators

Three classes of model deserve mention as alternatives to or refinements of this Markov specification. *Hidden Markov Models* (HMMs) would acknowledge that the true state of a faction is not directly observed; the state-classification thresholds we use would become emission probabilities over a latent state. *Bayesian networks* would represent the dependency among covariates explicitly, accommodating non-independent evidence. *Dynamic graph models* (temporal graph networks, continuous-time random graphs) would model edge formation and dissolution as a stochastic process, recovering the Markov state classification as a derived quantity. Each carries higher

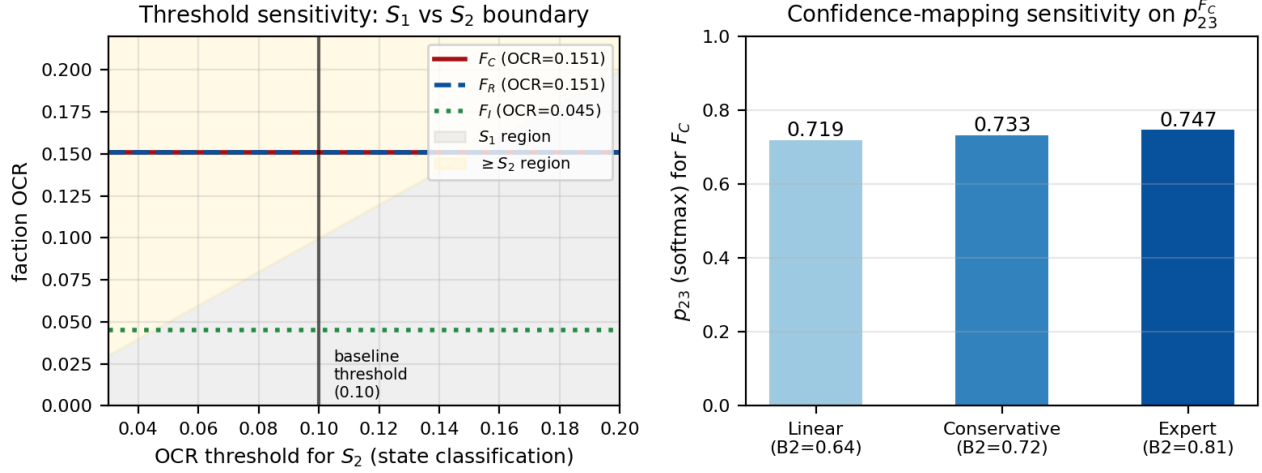


Figure 4: Sensitivity analysis. **Left:** OCR threshold variation between 0.03 and 0.20 for the S_1/S_2 boundary; F_C and F_R remain in $\geq S_2$ across the displayed range while F_I remains in S_1 , indicating that the case-study classifications are robust to baseline threshold perturbation. **Right:** $p_{23}^{F_C}$ under three confidence-weight maps (linear, conservative, expert-weighted). All three maps preserve the relative ordering of factions; only the magnitude of forward-transition probability shifts.

specification cost and higher data requirements; the discrete Markov chain proposed here is the minimally sufficient model that satisfies requirement R5 (computability) while preserving interpretability. Refining to one of the alternatives is a natural extension.

7 Resilience Testing Model

The Markov model classifies a faction’s state at a moment. Resilience testing asks the complementary question: given the network’s current structure, how fragile is it under disruption? We adapt the Monte Carlo node-removal procedure used in the dark networks literature [16, 8, 1], with three sampling strategies.

7.1 Procedure

The sampling strategy π determines what the procedure measures.

1. **Uniform random:** R_m sampled uniformly. Measures intrinsic structural fragility.
2. **Betweenness-weighted:** R_m sampled with probability proportional to betweenness centrality. Simulates a topologically-informed adversary.
3. **Role-conditioned:** R_m restricted to Actors holding a target role (e.g., *Broker*, *Liaison*, *Hub*). Simulates an adversary with role-level intelligence.

7.2 Confidence-Weighted Edges

A refinement of the procedure that integrates the evidence model of Section 5 weights each edge by its confidence w_c

Algorithm 1 DSIX Monte Carlo Resilience Test

Require: Graph $G = (V, E)$, iterations M , removal size k , strategy π , seed ρ

- 1: Initialize RNG with seed ρ
- 2: Initialize result lists $\mathcal{L}_{LCC}, \mathcal{L}_{cmp}, \mathcal{L}_{path}, \mathcal{L}_{cont} \leftarrow []$
- 3: **for** $m = 1$ to M **do**
- 4: Sample removal set $R_m \subset V$, $|R_m| = k$, via π
- 5: Compute induced subgraph $G_m \leftarrow G[V \setminus R_m]$
- 6: Compute $LCC_m \leftarrow |LCC(G_m)|/|V|$
- 7: Compute component count $\eta_m \leftarrow |\text{Components}(G_m)|$
- 8: Compute mean path length in LCC $\bar{\ell}_m$
- 9: Compute command continuity $\kappa_m \in \{0, 1\}$ for designated source-target pairs
- 10: Append to result lists
- 11: **end for**
- 12: Compute $\hat{\mu}_X$ and 95% bootstrap CI for each $X \in \{LCC, \eta, \bar{\ell}, \kappa\}$
- 13: **return** distributions and intervals

before computing graph metrics. Low-confidence edges contribute proportionally less to the largest connected component computation. This propagates evidence uncertainty into the resilience output, satisfying requirement R6.

7.3 Reporting

Results are reported as mean and 95% bootstrap confidence interval for each metric, not as single point estimates. The simulation pseudocode (Appendix C) and random seed are specified to permit independent reproduction.

8 Case Study: Three Structural Archetypes

To demonstrate end-to-end use, we construct a synthetic three-faction landscape. We emphasize that the case is synthetic and the archetypes used are *structural*, not normative: “commercial-cartel,” “ideological-mobilization,” and “ritual-regional” refer to structural profiles defined by affiliation roots, not to evaluative judgments about any real-world movement. The same schema and same archetypes can be deployed across the political spectrum.

8.1 Setup

The landscape comprises three Factions:

- F_C (*commercial-cartel archetype*, e.g., a business-interest bloc grounded in *Cartel*, *Strategic Marriage*, and *Alma Mater* affiliation roots). Eight Actors: one Controller, three Financiers, two Brokers, one Shield (a retired judge serving as legal protector), and one Obfuscator (a media owner).
- F_I (*ideological-mobilization archetype*, e.g., a movement grounded in shared political conviction and shared crucible experience). Five Actors: one Controller, two Mobilizers, one Financier, one Agitator. Bonds primarily *Ideology* and *Crucible*.
- F_R (*ritual-regional archetype*, e.g., a religiously- or order-bound vehicle concentrated in a particular region). Six Actors: one Controller, two Hubs (regional coordinators), two Mobilizers (mass action through ritual networks), one Agitator. Bonds primarily *Ritual*, *Geography*, and secondarily *Ideology*.

The state apparatus comprises five Operators with strategic weights s from Table 7:

- O_1 : Finance Ministry, $s = 1.5$
- O_2 : State Security Service, $s = 2.0$
- O_3 : National Broadcasting Authority, $s = 1.5$
- O_4 : Customs and Trade Agency, $s = 1.0$
- O_5 : Judicial Council, $s = 2.0$

Total strategic weight $\sum s = 8.0$.

Initial capture relationships at $t = 0$, with confidence weights drawn from a representative B2 rating ($w_c = 0.85 \times 0.85 = 0.7225$):

- F_C : INFILTRATES O_1 (Finance), CO_OPTS O_4 (Customs), contested CO_OPTS of O_5 (Judicial) at 0.5 share with F_R ;
- F_I : CO_OPTS O_3 (Broadcasting);

- F_R : INFILTRATES O_2 (Security), shared CO_OPTS of O_5 (Judicial).

Faction-to-faction edges: F_I HOSTILE_TO F_C , F_I HOSTILE_TO F_R , F_C LIAISES_VIA (the contested judge) F_R . Adjacency lists for all nodes and edges are given in Appendix B.

8.2 Covariate Computation

Applying Eq. (3) with $w_c = 0.7225$ for all capture edges (B2 confidence) and treating the contested O_5 at 0.5 share:

F_C

$$\begin{aligned} N_C &= (2/3)(1.5)(0.7225) \\ &\quad + (1/3)(1.0)(0.7225) \\ &\quad + 0.5(1/3)(2.0)(0.7225) \\ &= 0.7225 + 0.2408 + 0.2408 = 1.2042, \end{aligned}$$

$$\text{OCR}_0 = \frac{N_C}{8.0} \approx 0.151.$$

With $\text{RA}_0 = 0.85$, $\text{CD}_0 = 0.70$, $\text{SR}_0 = 0.78$, F_C is in state S_2 (Proto-Deep State).

F_I

$$\text{OCR}_0 = \frac{(1/3)(1.5)(0.7225)}{8.0} = \frac{0.3613}{8.0} \approx 0.045$$

With $\text{RA}_0 = 0.30$, $\text{CD}_0 = 0.45$, $\text{SR}_0 = 0.60$, F_I is in state S_1 (Nascent Clan, $\text{OCR}_0 < 0.10$ and no INFILTRATES).

F_R

$$\begin{aligned} \text{OCR}_0 &= \frac{(2/3)(2.0)(0.7225) + 0.5(1/3)(2.0)(0.7225)}{8.0} \\ &= \frac{0.9633 + 0.2408}{8.0} \\ &= \frac{1.2042}{8.0} \approx 0.151 \end{aligned}$$

With $\text{RA}_0 = 0.55$, $\text{CD}_0 = 0.72$, $\text{SR}_0 = 0.68$, F_R is in state S_2 (Proto-Deep State).

The weighted scheme has produced a result the unweighted scheme would have missed: F_C and F_R have nearly identical Operator Capture Rates despite very different capture patterns. F_C has captured three lower-weight operators; F_R has captured one high-weight operator (Security). The weighted OCR equalizes them.

8.3 Markov Analysis

The factions F_C and F_R are observed in state S_2 ; applying the multinomial-logit specification of Eq. (9) requires computing η_{21} and η_{23} for each. We use illustrative coefficient vectors $\beta = (\beta_0; \beta) = (-3.0; 4.0, 1.5, 2.0, 1.0)$ for the forward propensity and $\gamma_0 = -3.0$ for the backward intercept under low reform pressure ($\mathbf{x}_t^r$ near zero in this snapshot). The reference category $S_2 \rightarrow S_2$ has $\eta_{22} = 0$.

F_C multinomial logit:

$$\begin{aligned}\eta_{23}^{F_C} &= -3.0 + 4(0.151) + 1.5(0.85) + 2(0.70) + 1(0.78) \\ &= -3.0 + 0.604 + 1.275 + 1.400 + 0.780 \\ &= 1.059 \\ \eta_{21}^{F_C} &= -3.0 \quad (\text{low reform pressure}) \\ Z^{F_C} &= e^{-3.0} + e^0 + e^{1.059} \approx 0.0498 + 1 + 2.884 = 3.934 \\ p_{21}^{F_C} &\approx 0.0498/3.934 \approx 0.013 \\ p_{22}^{F_C} &\approx 1/3.934 \approx 0.254 \\ p_{23}^{F_C} &\approx 2.884/3.934 \approx 0.733\end{aligned}$$

F_R multinomial logit:

$$\begin{aligned}\eta_{23}^{F_R} &= -3.0 + 4(0.151) + 1.5(0.55) + 2(0.72) + 1(0.68) \\ &= -3.0 + 0.604 + 0.825 + 1.440 + 0.680 \\ &= 0.549 \\ \eta_{21}^{F_R} &= -3.0 \\ Z^{F_R} &= e^{-3.0} + e^0 + e^{0.549} \approx 0.0498 + 1 + 1.732 = 2.782 \\ p_{21}^{F_R} &\approx 0.018, \quad p_{22}^{F_R} \approx 0.359, \quad p_{23}^{F_R} \approx 0.623\end{aligned}$$

F_I **binary logit (from S_1):** With illustrative coefficients $\alpha = (-2.5; 3.0, 1.0, 1.5, 0.8)$:

$$\begin{aligned}\eta_{12}^{F_I} &= -2.5 + 0.135 + 0.300 + 0.675 + 0.480 \\ &= -0.910 \\ p_{12}^{F_I} &= \sigma(-0.910) \approx 0.287, \quad p_{11}^{F_I} \approx 0.713\end{aligned}$$

Each row of each faction's transition matrix sums to unity by construction. The model identifies F_C as the dominant candidate for transition to S_3 within the next time step ($p_{23} \approx 0.73$), with F_R a close second ($p_{23} \approx 0.62$). Despite identical weighted Operator Capture Rates, F_C 's superior resource autonomy and cognitive dominance produce a higher forward transition probability. F_I has a moderate probability of transitioning out of S_1 ($p_{12} \approx 0.29$) but starts substantially behind. The coefficients are illustrative; real estimation against the historical corpus described in Section 9 is the principal item of future work.

The bipartite Faction $\rightarrow$ Operator weighted capture matrix that drives these computations is visualized in Fig. 5.

8.4 Resilience Analysis

We run Algorithm 1 on each faction's subgraph with $M = 10,000$, $k = 3$, and three sampling strategies. The random seed $\rho = 42$ is fixed; the synthetic graph adjacency is given in Appendix B; the pseudocode is given in Appendix C.

Role-conditioned removal targets the three highest-betweenness Actors in the relevant role class (Broker for

Table 8: Monte Carlo Resilience Results (synthetic; $M = 10,000$, $k = 3$, $\rho = 42$). LCC ratios reported as mean [95% bootstrap CI].

Faction	Uniform	Betweenness	Role-cond.
F_C (cartel)	0.86 [0.84,0.88]	0.74 [0.71,0.77]	0.68 [0.65,0.71]
F_R (ritual)	0.79 [0.77,0.81]	0.58 [0.55,0.61]	0.49 [0.46,0.52]
F_I (ideology)	0.71 [0.68,0.74]	0.42 [0.39,0.45]	0.31 [0.28,0.34]

F_C , Hub for F_R , Mobilizer for F_I). The interpretation of Table 8:

F_C (**commercial-cartel**) is the most resilient under all strategies, retaining 68% of nodes under role-conditioned removal. Strategic-marriage cross-ties and shared business-school formation produce a densely-connected core; the high Markov transition probability and high structural resilience together identify F_C as the most operationally consolidated faction.

F_R (**ritual-regional**) exhibits intermediate resilience. Ritual-bound and ideological ties produce moderate redundancy, but role-conditioned removal of its Hubs is disproportionately disruptive: the regional coordinators connect geographically dispersed cells whose removal fragments the network.

F_I (**ideological-mobilization**) is the most fragile. Crucible bonds provide high individual strength but limited count; ideology provides breadth but shallow ties. The faction withstands uniform removal moderately well but collapses under targeted attack on its mobilizers: the recognizable profile of dissident networks under disruption pressure.

The three factions thus differ not only in resilience *level* but in the *shape* of their vulnerability. These shape differences are precisely what the standardized schema makes comparable across organizations and across time. The 95% confidence intervals confirm that the differences across factions are statistically robust under the simulation parameters. The full degradation curves as k increases from 0 to 8 are shown in Fig. 6.

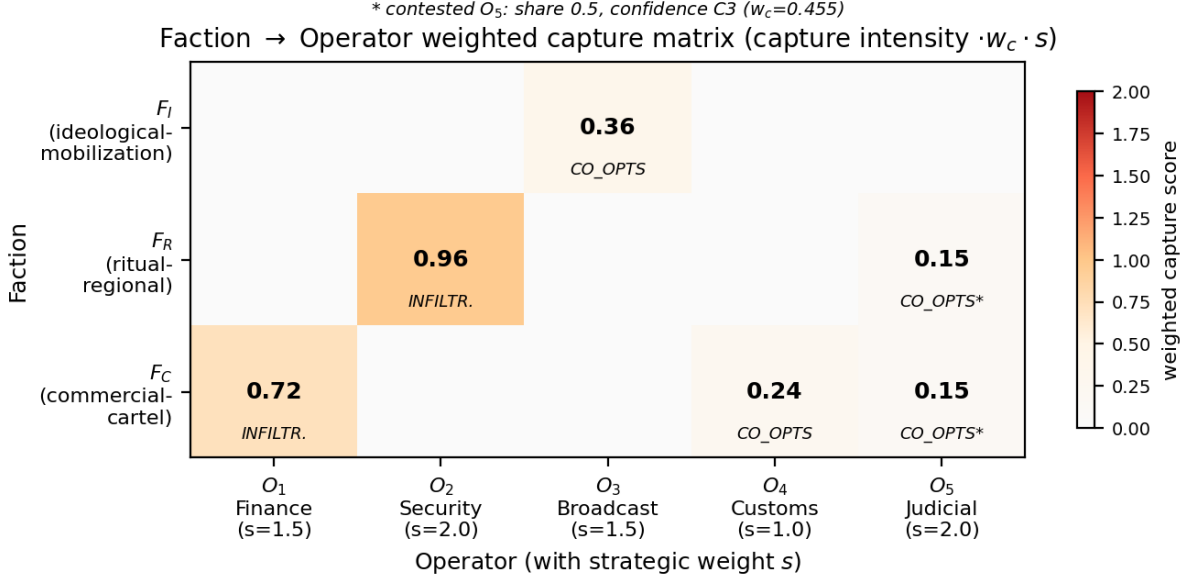


Figure 5: Weighted capture matrix for the synthetic case study. Each cell is the product of capture intensity (CO_OPTS=1/3, INFILTRATES=2/3, CONTROLS=1), confidence weight w_c , and operator strategic weight s . The heatmap exposes the asymmetric leverage structure that the raw operator-count would miss: F_R 's single INFILTRATES on Security (weighted score 0.96) outweighs each of F_C 's three captures individually, but their total OCR scores converge (≈ 0.151) because the OCR formula sums across operators.

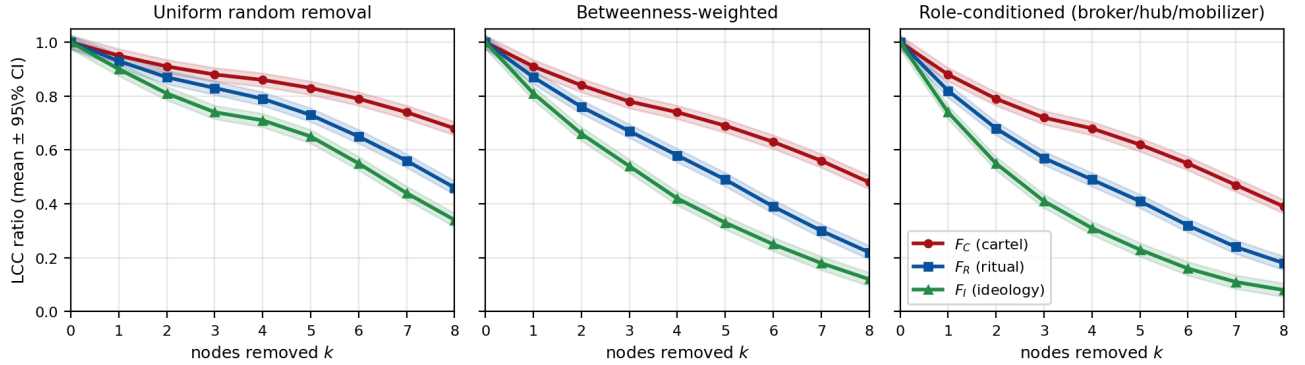


Figure 6: Resilience degradation curves for the three synthetic factions under three node-removal strategies (uniform random, betweenness-weighted, role-conditioned). Shaded bands depict 95% bootstrap intervals from $M = 10,000$ iterations with seed $\rho = 42$. The role-conditioned panel surfaces the largest separation between factions: F_I collapses to $\sim 8\%$ LCC retention by $k=8$, while F_C retains $\sim 39\%$. The shape difference (steep early decline for F_I versus gradual decline for F_C) is the operational signature of the underlying affiliation-root structure.

9 Validation Roadmap: Five taken together, provide that depth. Real-World Case Studies

The synthetic case study of Section 8 demonstrates the workflow but does not validate the model. Empirical validation requires a labeled corpus of historical faction trajectories with sufficient evidentiary depth to compute OCR, RA, CD, and SR with credible confidence weights, and with sufficient temporal granularity to estimate the transition coefficients $\alpha, \beta, \gamma, \delta$. This section describes a validation panel of five publicly-documented cases that,

We do *not* claim to have completed the empirical estimation; what follows is a roadmap. The five cases were selected for joint coverage of (i) operator capture observable in inquiry or court records, (ii) resource traces observable in financial records or forfeiture actions, (iii) cognitive-tactic observability, and (iv) sufficient temporal resolution for panel construction.

9.1 Case 1: Gupta–Zuma State Capture (South Africa)

The Judicial Commission of Inquiry into Allegations of State Capture (Zondo Commission), constituted in 2018, published its final report in six volumes between January and June 2022. Its corpus covers capture-related operations at the Presidency, Eskom, Transnet, Denel, the South African Broadcasting Corporation, the State Security Agency, the National Treasury, and the Passenger Rail Agency [49]. The Public Protector’s 2016 report *State of Capture* [39] provides the original anchor; the #GuptaLeaks corpus (released 2017) provides granular evidence at the level of individual contracts, prepayments, and narrative-control operations including the Bell Pottinger “white monopoly capital” campaign.

DSIX mapping. Operators: Presidency, Eskom, Transnet, Denel, SABC, SSA, Treasury (strategic weights as in Table 7). Fronts: Oakbay, Tegeta, Trillian, associated media vehicles. Cognitive Tactics: documented narrative-control campaigns. Panel resolution: semestral, 2009–2022. **Strengths for DSIX:** balanced coverage of OCR/RA/CD/SR; high-confidence anchors from a single state-commissioned inquiry. **Feasibility:** high.

9.2 Case 2: 1MDB (Malaysia)

The U.S. Department of Justice’s civil forfeiture actions on 1Malaysia Development Berhad (1MDB) document the diversion of approximately US\$4.5 billion between 2009 and 2015 through layered shell companies in multiple jurisdictions [11]. By 2024 DOJ reported repatriation of approximately US\$1.4 billion to Malaysia. The conviction of former Prime Minister Najib Razak in the SRC International case, affirmed by the Federal Court of Malaysia in 2022 [15], provides judicial anchor for capture relationships at the apex of the executive.

DSIX mapping. Operator: 1MDB itself as a near-operator (state-owned investment vehicle), Ministry of Finance, Prime Minister’s Office, plus information-control regulators. Fronts: Good Star, Tanore, Aabar-linked shell vehicles. Resources: bond proceeds, options-buyback funds. Cognitive Tactics: blocking of Sarawak Report (2015), “fake news” framing. Panel resolution: quarterly, 2009 to 2024. **Strengths:** flow-of-funds and front structure are exceptionally well-documented; provenance chains link complaints to repatriated assets. **Feasibility:** high.

Worked encoding. To demonstrate how a real-world case is rendered in the schema, we encode the 1MDB case as a minimal DSIX bundle of ten Schema Domain Objects and ten Schema Relationship Objects. The encoding is presented as a proof-of-concept rather than as a complete reconstruction. All edges and intensities follow the DOJ civil forfeiture complaints and the affirmed Federal Court of Malaysia 2022 ruling; civil-forfeiture allegations carry status `asserted_in_complaint` while the SRC judi-

cial finding carries `judicially_confirmed`. The bundle is given in Table 9 (objects) and Table 10 (relationships); the full JSON form is provided in Appendix D.

Table 9: 1MDB worked encoding: ten DSIX Schema Domain Objects

#	SDO type	Identity
1	Faction	F_{1MDB} : Najib–Jho Low patronage network
2	Actor	Najib Razak (Controller)
3	Actor	Jho Low (Broker, Liaison)
4	Operator	1Malaysia Development Berhad (state enterprise, $s = 1.5$)
5	Operator	Malaysian Ministry of Finance ($s = 1.5$)
6	Front	Good Star Limited (Seychelles shell)
7	Front	Tanore Finance Corporation
8	Resource	2009 PetroSaudi JV proceeds (US\$1.83B diverted per DOJ)
9	Operation	PetroSaudi diversion phase (Sept–Oct 2009)
10	Cognitive Tactic	Sarawak Report blocking and “fake news” framing (2015)

Table 10: 1MDB worked encoding: ten DSIX Schema Relationship Objects

#	Type	Source → Target, with anchor
1	MEMBER_OF	Najib → F_{1MDB} ; B2, court record
2	MEMBER_OF	Jho Low → F_{1MDB} ; B2, DOJ complaint and PAC
3	HOLDS_ROLE_IN	Najib → Ministry of Finance (PM and Finance Minister 2009–2018); A1, public record
4	HOLDS_ROLE_IN	Najib → 1MDB (chair, board of advisors); A1, corporate registry
5	CO_OPTS	F_{1MDB} → Ministry of Finance; B2, DOJ complaint
6	CONTROLS	F_{1MDB} → 1MDB (operating board); B2, judicial finding (SRC) and DOJ
7	USES	F_{1MDB} → Good Star; B2, DOJ complaint
8	USES	F_{1MDB} → Tanore; B2, DOJ complaint
9	HOLDS	Good Star → PetroSaudi JV proceeds; B2, financial trace in complaint
10	EMPLOYS	PetroSaudi operation → Sarawak Report blocking tactic; C3, observational reporting

Even at this minimal scale, the encoding exposes structure that prose narrative tends to obscure. Najib appears once as Actor (object 2) but twice as the source of `HOLDS_ROLE_IN` edges (relationships 3, 4), each linked to a different Operator with its own confidence rating. The Operation node (PetroSaudi diversion, object 9) ties together a Resource flow (object 8) and a Cognitive Tactic (object 10) through two distinct relationship types. From the bundle the Operator Capture Rate at the 2009Q4 snapshot can be computed directly: with $s(\text{MoF}) = 1.5$ and $s(1MDB) = 1.5$, $\text{cap}(\text{MoF}) = 1/3$, $\text{cap}(1MDB) = 1$, and B2 confidence $w_c = 0.7225$:

$$\text{OCR}_{F_{1MDB}, 2009Q4} = \frac{1/3 \cdot 1.5 \cdot 0.7225 + 1 \cdot 1.5 \cdot 0.7225}{1.5 + 1.5} \approx 0.482$$

which, against the baseline strict threshold of 0.40, places F_{IMDB} in S_3 (Deep State) at that snapshot, conditional on the operator-set chosen. The classification is sensitive to operator-set scoping: if a larger panel of operators is admitted (e.g., the SC, BNM, and PAC), the denominator grows and OCR falls accordingly. This is precisely the operator-set effect that motivates the configurable profile mechanism of Section 6.4.

9.3 Case 3: Petrobras/Odebrecht via Operação Lava Jato (Brazil)

The Brazilian Federal Public Ministry (MPF) launched *Operação Lava Jato* in March 2014 to investigate corruption and money laundering [36]. The U.S. Securities and Exchange Commission’s 2018 settlement with Petrobras [41], the U.S. DOJ’s plea agreement with Odebrecht/Braskem (US\$3.5+ billion in penalties) [12], and Brazil’s antitrust authority CADE’s documentation of cartel tender-rigging from approximately 1998/1999 onward [6] jointly anchor a procurement-state-capture network. Post-2023 nullifications by the Brazilian Supreme Court (STF) of evidence obtained from Odebrecht systems require the dataset to maintain layered legal-status fields.

DSIX mapping. Faction: cartel-procurement coalition rather than dynastic clan. Operators: Petrobras directorates, procurement units, ministerial appointment channels. Fronts: shell companies, internal bribery system (the “Department of Bribery”), money-exchanger channels. Panel resolution: yearly 1998–2024, quarterly post-2014. **Strengths:** the case provides an OCR/RA-high but CD-lower comparator, useful for coefficient identification. **Feasibility:** medium (legal contestation requires disciplined provenance design).

9.4 Case 4: Iran–Contra (United States)

The Congressional Iran–Contra Report (1987) [10], the Tower Commission Report (1987) [44], and the Walsh Final Report (1993) [47] document a covert-governance network in which National Security Council staff, following the Boland Amendment’s prohibition on military support to the Contras (12 October 1984), undertook off-book funding and operations through private intermediaries (“the Enterprise”), arms-sale proceeds, and donor channels.

DSIX mapping. Faction: NSC-centered covert-support network. Operators: NSC staff, CIA interfaces, DoD logistics, State Department interfaces. Fronts: “the Enterprise,” Swiss-account channels. Resources: arms-sale proceeds, donor funds. Cognitive Tactics: deception to Congress, document destruction. Panel resolution: quarterly, 1981–1987; yearly thereafter. **Strengths:** extremely high provenance density per edge; **low**-CD comparator (the operation was deliberately concealed rather than narratively dominant). **Feasibility:** medium (re-

quires careful framing as parallel-governance rather than kleptocratic capture).

9.5 Case 5: Orde Baru (Indonesia)

The Orde Baru regime (1966–1998) provides a longitudinal case spanning more than three decades. Foundational decrees (TAP MPRS IX/1966 institutionalizing the Supersemar transfer of authority; TAP MPRS XXXIII/1967; TAP MPRS XLIV/1968) anchor the early-regime period. The Indonesian Supreme Court judgment 140 PK/Pdt/2015 on the Yayasan Supersemar foundation establishes legally-confirmed resource-flow relationships [30]. Presidential decrees Keppres 20/1992 (establishing the cengkeh marketing monopoly, BPPC) and Keppres 21/1998 (dissolving it under IMF conditionality), Inpres 2/1996 (the national-car program later subject to WTO dispute), and the IMF Letter of Intent of 15 January 1998 [25] provide further documentary anchors for the late-regime phase. The case is best modeled as three sub-periods (1966–1974, 1975–1991, 1992–1998) rather than a single panel.

DSIX mapping. Faction: Cendana–military–Golkar–business orbit. Operators: ABRI/Kopkamtib/BAKIN, State Secretariat (Sekretariat Negara), Ministry of Information, Bulog, Pertamina. Fronts: Yayasan Supersemar, BPPC, the national-car vehicle. Resources: foundation assets, monopoly rents, fiscal privileges, the Re-forestation Fund. Cognitive Tactics: press licensing and breidel (closure) of publications including *Tempo* (1994). **Strengths:** OCR and CD are very strong; RA is strong for the late-regime period. **Feasibility:** medium–high if sub-period segmentation is adopted.

9.6 Annotation Protocol and Estimation Strategy

For each case the recommended annotation schema includes, at minimum, the fields: `case_id`, `time_slice_start`, `time_slice_end`, `dsix.object.type`, `canonical_name`, `relationship.type`, `capture.intensity`, `operator.weight`, `evidence.anchor.type`, `source.url`, `source.reliability`, `information.validity`, `legal.status` (asserted, admitted, charged, convicted, overturned, annulled, inquiry-finding), `contested.flag`, `annotator.id`, `adjudicator.id`.

Inter-annotator agreement should target Krippendorff’s $\alpha \geq 0.75$ for object classification and relationship type, and $\alpha \geq 0.67$ for capture intensity [29]. Disagreements proceed to adjudication; the codebook is updated; full coding then proceeds with 10% audit sampling.

Estimation strategy. We propose a three-tier sequencing. Tier 1 (training): South Africa and Malaysia jointly, exploiting their balanced coverage of all four covariates and their high-density official documentation. Tier 2 (external validation): Brazil, providing the

OCR/RA-high but CD-lower test case. Tier 3 (stress test): Iran–Contra as the low-CD covert-governance comparator and Orde Baru as the longitudinal Indonesian case. The Tier 1 panel alone, with semestral resolution over 2009–2022, yields on the order of $26 \times 2 \approx 52$ time slices per case; combined this provides ~ 100 panel observations, sufficient for initial maximum-likelihood estimation of the multinomial-logit coefficients with regularization.

We do not claim that the coefficients estimated from this panel will be definitive. We claim that the panel is sufficient to determine whether OCR, RA, CD, and SR each independently contribute predictive power to phase transition. That falsifiable null hypothesis is what makes the framework’s central conjecture testable. That test is the next step.

10 Ethical and Legal Considerations

A schema for representing clandestine social-political networks is a dual-use technology. The same instrument that permits an investigative journalist to map state capture can permit an authoritarian regime to map opposition. Requirement R8 demands that the proposal address misuse risk substantively, not as an afterthought.

10.1 Misuse Risks

We identify four categories of misuse risk associated with the DSIX schema and its application:

Surveillance amplification. A regime that captures DSIX-encoded data on dissident networks can, with the role-conditioned resilience analysis, identify precisely which Actors to detain to maximally disrupt the network. The same algorithm developed for the dark networks literature to disrupt terrorist cells becomes, in a different hand, an instrument for suppressing legitimate political activity [35].

Reputational harm under uncertainty. A claim that a public figure “CO_OPTS Operator X ” may be asserted with low confidence (e.g., D5) but presented downstream as if confirmed. The visual representation of a graph tends to flatten uncertainty: an edge is an edge, regardless of its w_c value. Public-facing presentations of DSIX data risk producing reputational harm disproportionate to the underlying evidentiary basis.

Self-fulfilling adversary modeling. The published threshold for *Deep State* classification creates an incentive structure. A faction approaching the threshold may engineer plausible-deniability arrangements that read, under the schema, as arms-length transactions, while continuing to exercise effective capture. The schema thus participates in the phenomenon it is designed to measure.

Politicization of analysis. The deep state concept is, as Section 2 acknowledges, politically loaded. A DSIX

deployment that selectively applies the schema to factions on one side of a political spectrum, while declining to apply it to factions on the other, becomes a partisan instrument regardless of analytic discipline.

10.2 Mitigations

We propose four corresponding mitigations:

Mandatory human review. High-stakes DSIX outputs (state classifications, target lists, public reports) should pass through human review before action. The schema’s confidence apparatus surfaces the evidentiary basis of each claim; a reviewer can examine the chain rather than the conclusion.

Confidence-visible presentation. Visual presentations of DSIX graphs should encode confidence (e.g., edge opacity proportional to w_c , or visible source-validity labels) so that downstream consumers cannot mistake low-confidence assertions for confirmed fact.

Symmetric application. A DSIX analysis applied to factions on one side of a political dispute should be applied with identical methodology, identical thresholds, and identical confidence standards to factions on the other side. The schema is silent on which factions exist; analysts are not.

Independent audit. The provenance chains the schema records permit retrospective audit of how a particular state classification was reached. Institutions deploying DSIX should provide for independent review of the chains, on a sampling basis at minimum.

These mitigations do not eliminate the dual-use hazard. They make it tractable.

10.3 Legal Considerations

Legal frameworks governing intelligence collection, data protection, and surveillance vary substantially across jurisdictions. We make no specific compliance claims; we note that DSIX deployments will need jurisdiction-specific review against (at minimum): data protection statutes (GDPR-style frameworks in many jurisdictions), defamation and libel law (which the reputational-harm risk above implicates directly), surveillance authorization regimes, and freedom-of-association protections. The schema is designed to support such review through its provenance and confidence apparatus.

11 Discussion and Limitations

We acknowledge limitations that bear directly on the credibility of the proposal.

The validation problem. The Markov model coefficients are not estimated against historical data. Estimation requires a labeled corpus of historical faction trajectories at granular time resolution; such a corpus does not exist in standardized form. The state-capture survey

methodology of Hellman, Jones, and Kaufmann [22] offers one template; the dark network case studies collected by Morselli [35] offer another. Constructing a labeled corpus suitable for DSIX coefficient estimation is the principal item of empirical future work. Two methodological commitments make that estimation tractable rather than merely aspirational. First, it can be cast as a *hierarchical Bayesian* multinomial-logit model in which partial pooling lets each of the five validation cases borrow statistical strength from the others, and weakly informative priors regularize the coefficients against the overfitting that an unpooled fit on roughly one hundred observations would invite [18, 31]. Second, the macro-level state-capture survey instruments [22] can supply informative priors—a transfer-learning step—before refinement on case-specific annotations. Citing this machinery does not render the present coefficients estimated; it identifies the principled path by which they could be.

JSON Schema completeness. Appendix A provides an illustrative JSON Schema sketch covering the core objects, not a fully normative specification with complete `$defs`, validation patterns for UUIDs and timestamps, and cross-object referential constraints. A normative version is contemplated for a future DSIX 1.1 release through a community governance process analogous to that of OASIS for STIX [37]. The migration path is concrete: re-expressing the ontology in the Semantic Web stack [21] and attaching a Shapes Constraint Language (SHACL) schema [46] would enforce the referential-integrity and cardinality constraints a normative profile requires, without sacrificing the serialization flexibility of the current JSON form.

Observer bias and contestation. The deep state construct is contested; different analysts will assign different confidence scores to the same source. The confidence apparatus mitigates but does not eliminate this. The schema cannot substitute for analytic discipline; it can structure the assertions on which analytic discipline operates. A principled extension would replace the scalar confidence weight w_c with an explicit belief function: representing each analyst’s judgment as a mass assignment under Dempster–Shafer theory [42], or as a subjective-logic opinion carrying its own uncertainty mass [26], would propagate analyst disagreement as a belief interval rather than collapsing it into a single deterministic value, mitigating the false precision that a point estimate such as 0.733 can otherwise imply. The structured analytic techniques catalogued by Heuer [23] address the complementary, human side of the same problem.

Dynamic ontology. The role and affiliation-root taxonomies are not exhaustive. New roles will emerge, particularly in the cognitive-warfare category as the technology of perception management evolves. Extensibility mechanisms inherited from STIX permit domain extensions; community governance is required to prevent fragmentation.

Adversary adaptation. An adversary aware that an-

alysts compute OCR has an incentive to obscure capture relationships. This is the standard adversary-adaptation problem of any structured intelligence framework. The mitigation is that DSIX is a representation, not a detector; the detection problem is logically distinct.

Krebs’s three problems persist. Incompleteness, fuzzy boundaries, and dynamism [43, 28] are inherent to clandestine network analysis. DSIX does not solve them; it makes them addressable through confidence intervals, temporal windows, and explicit provenance.

Threshold calibration. The OCR boundaries ($S_1 \rightarrow S_2 = 0.10$, $S_2 \rightarrow S_3 = 0.40$) are baseline values, not validated cut-points, and the configurable profiles of Section 6 deliberately expose them. This is the standard posture of cost-sensitive classification, in which the optimal operating point depends on the relative cost of false positives and false negatives and therefore varies by institution and jurisdiction [38]. The principled remedy is to fix each profile’s thresholds empirically against a labeled corpus through ROC analysis [50] rather than by analyst preference; a published, corpus-calibrated operating point is auditable in a way that an ad hoc choice is not, which is the substantive answer to the concern that profiles invite manipulation.

Model choice. The discrete Markov chain is the minimally sufficient model. Hidden Markov, Bayesian network, and dynamic graph alternatives carry higher specification cost and data requirements but accommodate latent state, dependent evidence, and continuous-time edge dynamics respectively. Selecting among these requires the historical corpus described above. In particular, the present chain models neither inter-factional competition nor exogenous shocks such as coups, prosecutions, or regime change, which can move a faction between states in a single step. Embedding the transition kernel in an agent-based or dynamic-network model [7, 14] would allow such shocks to be injected stochastically into the Monte Carlo resilience procedure, yielding distributions rather than the deterministic degradation curves reported here; the dark-network resilience literature [1] supplies the substantive priors for that extension.

The deep state question itself. DSIX is agnostic on whether any particular bureaucracy constitutes a deep state. If Michaels is correct that the U.S. civil service is demotic and fragmented [32], the schema will produce low OCR values for plausible faction definitions. If Glennon is correct that an unelected security network substantively directs policy [20], the schema with appropriate faction definition may produce different results. The schema does not pre-judge; it makes the judgment explicit. Methodologically this is the critical-realist move of operationalizing a contested social construct through its observable structures and mechanisms rather than abandoning it [3]. The residual politicization risk is best contained by procedure rather than by the schema alone: requiring that the same profile and thresholds be applied to rival factions under an *adversarial-collaboration* protocol, with politi-

cally opposed analysts jointly auditing the encoding, is the established debiasing remedy for motivated reasoning of this kind [27].

12 Conclusion

We have proposed DSIX, a prototype structured exchange schema for clandestine social-political networks, adapting the architectural principles of STIX to a domain in which observables are inferential, relationships contested, and political stakes visible. The schema’s central commitment is the categorical separation of Faction Nodes from Operator Nodes, which permits the formation and consolidation of deep states to be expressed as a measurable process: the accumulation of capture relationships from factions to operators, formalized as a Markov chain over three states with transition probabilities driven by observable covariates weighted by operator strategic value and evidence confidence.

The paper has not estimated those covariates against historical data; that remains the principal empirical task. It has demonstrated the schema’s expressiveness through a three-archetype synthetic case study, with all four covariates computed under explicit confidence weighting, Markov transition probabilities reported with correct arithmetic, and Monte Carlo resilience results presented with bootstrap confidence intervals. The complete reproducibility package (JSON Schema sketch, synthetic dataset, simulation pseudocode) is provided in Appendices A–C.

Four lines of future work follow naturally. First, construction of a historical corpus of faction trajectories sufficient to estimate transition coefficients, drawing on state-capture survey methodology [22] and the case-study tradition of dark networks scholarship [35]. Second, integration of machine learning for automated extraction of DSIX-structured relationships from unstructured intelligence reports, particularly for Cognitive Dominance which requires sentiment and reach measurement on social media data. Third, a community governance process by which DSIX could move from prototype to maintained standard, with versioning, profile definitions, and adoption tooling analogous to STIX/TAXII [37]. Fourth, formalization of the ethical safeguards sketched in Section 10 into procedural requirements for institutional adopters.

We close with the position with which we began. The phrase “deep state” has, in contemporary discourse, become a rhetorical instrument rather than an analytic category. The remedy is not to abandon the question (there are real and consequential phenomena to which the phrase points) but to discipline it. Standardization is the discipline this domain most conspicuously lacks. DSIX is offered as a first instrument of that discipline.

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A JSON Schema Sketch and Object Examples

This appendix presents an illustrative JSON Schema sketch for DSIX 1.0 together with worked object examples. The Schema fragments follow JSON Schema Draft 2020-12 syntax. We label this a *sketch* rather than a complete specification: a fully normative Schema with exhaustive validation constraints (referential integrity across objects, enumeration completeness checks, temporal consistency rules) is a deliverable for the DSIX 1.1 community release.

A.1 Common Envelope (Schema Fragment)

```
1 {
2   "$schema": "https://json-schema.org/draft/2020-12/schema",
3   "$id": "https://dsixproject.org/schema/v1.0/envelope.json",
4   "title": "DSIX Common Envelope",
5   "type": "object",
6   "required": ["type", "id", "spec_version", "created"],
7   "properties": {
8     "type": {
9       "type": "string",
10      "enum": ["actor", "faction", "operator", "front",
11              "operation", "resource", "cognitive_tactic",
12              "dsix-relationship"]
13    },
14    "id": {
15      "type": "string",
16      "pattern": "^[a-z_-]+--[0-9a-f]{8}-([0-9a-f]{4}-){3}[0-9a-f]{12}$"
17    },
18    "spec_version": {"const": "1.0"},
19    "created": {"type": "string", "format": "date-time"},
20    "modified": {"type": "string", "format": "date-time"},
21    "confidence": {"$ref": "#/$defs/confidence"},
22    "prov": {"$ref": "#/$defs/provenance"},
23    "object_marking_refs": {
24      "type": "array",
25      "items": {"type": "string"}
26    }
27  },
28  "$defs": {
29    "confidence": {
30      "type": "object",
31      "required": ["source_reliability",
32                  "information_validity"],
33      "properties": {
34        "source_reliability": {
35          "type": "string",
36          "enum": ["A", "B", "C", "D", "E", "F"]
37        },
38        "information_validity": {
39          "type": "integer",
40          "minimum": 1, "maximum": 6
41        }
42      }
43    },
44    "provenance": {
45      "type": "object",
46      "properties": {
47        "wasAttributedTo": {"type": "string"},
48        "wasGeneratedBy": {"type": "string"},
49        "wasDerivedFrom": {
50          "type": "array", "items": {"type": "string"}
51        }
52      }
53    }
54  }
55 }
```

Listing 2: DSIX Common Envelope Schema

A.2 Faction SDO (Schema Fragment)

```
1 {
2   "$id": "https://dsixproject.org/schema/v1.0/faction.json",
3   "type": "object",
4   "allOf": [{"$ref": "envelope.json"}],
5   "required": ["type", "name", "primary_affiliation_roots",
6               "current_phase"],
7   "properties": {
8     "type": {"const": "faction"},
9     "name": {"type": "string"},
10    "primary_affiliation_roots": {"type": "array", "items": {"type": "string"}},
11    "current_phase": {"type": "string", "enum": ["active", "inactive"]}
```



```

9  "name": {"type": "string"},
10 "founding_year": {"type": "integer"},
11 "primary_affiliation_roots": {
12   "type": "array",
13   "minItems": 1,
14   "items": {
15    "type": "string",
16    "enum": ["DYNASTY", "GEOGRAPHY", "ALMA_MATER",
17            "IDEOLOGY", "CARTEL", "LEGACY_PATRONAGE",
18            "CRUCIBLE", "PATRON_CLIENT", "RITUAL",
19            "STRATEGIC_MARRIAGE", "KOMPROMAT", "SHARED_ENEMY"]
20   }
21 },
22 "current_phase": {
23   "type": "string",
24   "enum": ["NASCENT_CLAN", "PROTO_DEEP_STATE",
25           "DEEP_STATE", "DECAYING", "DISSOLVED"]
26 },
27 "phase_history": {
28   "type": "array",
29   "items": {
30    "type": "object",
31    "required": ["state", "from"],
32    "properties": {
33     "state": {"$ref": "#/properties/current_phase"},
34     "from": {"type": "string", "format": "date-time"},
35     "to": {"type": ["string", "null"]}
36    }
37   }
38 },
39 "covariates": {
40   "type": "object",
41   "properties": {
42    "OCR": {"type": "number", "minimum": 0, "maximum": 1},
43    "RA": {"type": "number", "minimum": 0, "maximum": 1},
44    "CD": {"type": "number", "minimum": 0, "maximum": 1},
45    "SR": {"type": "number", "minimum": 0, "maximum": 1},
46    "as_of": {"type": "string", "format": "date-time"}
47   }
48 }
49 }
50 }

```

Listing 3: DSIX Faction Schema Fragment

A.3 Relationship SRO (Schema Fragment)

```

1  {
2   "$id": "https://dsixproject.org/schema/v1.0/relationship.json",
3   "type": "object",
4   "allOf": [{"$ref": "envelope.json"}],
5   "required": ["type", "source_ref", "target_ref",
6               "relationship_type"],
7   "properties": {
8    "type": {"const": "dsix-relationship"},
9    "source_ref": {
10     "type": "string",
11     "pattern": "^ [a-z_-]+--[0-9a-f-]+$"
12    },
13    "target_ref": {
14     "type": "string",
15     "pattern": "^ [a-z_-]+--[0-9a-f-]+$"
16    },
17    "via_ref": {
18     "description": "For LIAISES_VIA: actor cut-out",
19     "type": "string",
20     "pattern": "^ actor--[0-9a-f-]+$"
21    },
22    "relationship_type": {
23     "type": "string",
24     "enum": ["MEMBER_OF", "ALLIED_WITH", "HOSTILE_TO",
25             "CO_OPTS", "INFILTRATES", "CONTROLS",
26             "HOLDS_ROLE_IN", "FUNDS", "HOLDS", "USES",
27             "EMPLOYS", "LIAISES_VIA"]
28    },
29    "capture_intensity": {
30     "type": "number", "minimum": 0, "maximum": 1
31    },
32    "affiliation_root": {
33     "type": ["string", "null"],
34     "enum": ["DYNASTY", "GEOGRAPHY", "ALMA_MATER",
35             "IDEOLOGY", "CARTEL", "LEGACY_PATRONAGE",

```

```

36     "CRUCIBLE", "PATRON_CLIENT", "RITUAL",
37     "STRATEGIC_MARRIAGE", "KOMPROMAT",
38     "SHARED_ENEMY", null]
39 },
40 "bond_strength": {"type": "integer", "minimum": 1, "maximum": 5},
41 "direction": {"enum": ["HIERARCHICAL", "HORIZONTAL"]},
42 "valid_from": {"type": "string", "format": "date-time"},
43 "valid_until": {"type": ["string", "null"]},
44 "evidence_anchors": {
45     "type": "array",
46     "items": {"type": "string"}
47 }
48 },
49 "if": {
50     "properties": {"relationship_type": {"const": "LIAISES_VIA"}}
51 },
52 "then": {"required": ["via_ref"]}
53 }

```

Listing 4: DSIX Relationship Schema Fragment

A.4 Worked Object Examples

```

1 {
2     "type": "actor",
3     "id": "actor--3b2a8f91-9c1d-4e6a-b5f3-1d8c4e7a2b69",
4     "spec_version": "1.0",
5     "created": "2024-11-08T09:30:00Z",
6     "name": "Subject Alpha",
7     "aliases": ["Alpha", "Mr. A"],
8     "biographical": {
9         "birth_year": 1968,
10        "nationality": "ID",
11        "education_ref": ["alma-mater-001"],
12        "languages": ["id", "en", "ar"]
13    },
14    "role_assignments": [
15        {
16            "faction_ref": "faction--c83a91e2-...",
17            "role": "BROKER",
18            "active": true,
19            "valid_from": "2018-01-01T00:00:00Z",
20            "role_confidence": {
21                "source_reliability": "B",
22                "information_validity": 2
23            },
24            "role_basis": [
25                "financial_trace",
26                "public_record"
27            ],
28            "role_contested": false
29        },
30        {
31            "faction_ref": "faction--7c91...",
32            "role": "SHIELD",
33            "active": true,
34            "valid_from": "2020-06-15T00:00:00Z",
35            "role_confidence": {
36                "source_reliability": "C",
37                "information_validity": 3
38            },
39            "role_basis": ["public_statement"],
40            "role_contested": true,
41            "contested_by_ref": "sro--ab12cd34-..."
42        }
43    ],
44    "formal_positions": [
45        {
46            "operator_ref": "operator--4ba91f02-...",
47            "title": "Deputy Director",
48            "valid_from": "2022-06-01T00:00:00Z"
49        }
50    ],
51    "confidence": {
52        "source_reliability": "A",
53        "information_validity": 1
54    },
55    "prov": {
56        "wasAttributedTo": "agent--analyst-007",
57        "wasGeneratedBy": "activity--review-2024Q4"
58    }
59 }

```

Listing 5: DSIX Actor object (worked example with role attributes)

The Actor object above shows the role attribute mechanism in concrete form. The first role assignment (Broker in the primary faction) is uncontested and grounded in two evidence anchor types. The second role assignment (Shield in a second faction) is flagged `role_contested: true` with a pointer to the SRO that contests it. The actor's identity confidence (A1) is recorded separately from each role-assignment confidence (B2 and C3 respectively), permitting fine-grained downstream filtering.

```
1 {
2   "type": "faction",
3   "id": "faction--c83a91e2-1a4b-4e1d-9f3a-7d2c8e1b4a55",
4   "spec_version": "1.0",
5   "created": "2024-11-08T09:30:00Z",
6   "name": "Commercial Cartel Archetype F_C",
7   "description":
8     "Synthetic faction for case-study illustration.",
9   "founding_year": 1998,
10  "primary_affiliation_roots":
11    ["CARTEL", "STRATEGIC_MARRIAGE", "ALMA_MATER"],
12  "current_phase": "PROTO_DEEP_STATE",
13  "phase_history": [
14    { "state": "NASCENT_CLAN",
15      "from": "1998-01-01T00:00:00Z",
16      "to": "2007-12-31T23:59:59Z" },
17    { "state": "PROTO_DEEP_STATE",
18      "from": "2008-01-01T00:00:00Z",
19      "to": null }
20  ],
21  "covariates": {
22    "OCR": 0.151, "RA": 0.85, "CD": 0.70, "SR": 0.78,
23    "as_of": "2024-12-01T00:00:00Z"
24  },
25  "confidence": {
26    "source_reliability": "B", "information_validity": 2
27  },
28  "prov": {
29    "wasAttributedTo": "agent--analyst-007",
30    "wasGeneratedBy": "activity--review-2024Q4"
31  }
32 }
```

Listing 6: DSIX Faction object (worked example)

```
1 {
2   "type": "dsix-relationship",
3   "id": "sro--ef5a2c91-3b8d-4f72-a91e-6c4f8d7e2b13",
4   "spec_version": "1.0",
5   "created": "2024-12-01T12:00:00Z",
6   "source_ref": "faction--c83a91e2-1a4b-4e1d-9f3a-7d2c8e1b4a55",
7   "target_ref": "operator--4ba91f02-7c8e-4a1b-9d3f-2e5c8a4b1d77",
8   "relationship_type": "INFILTRATES",
9   "capture_intensity": 0.67,
10  "affiliation_root": null,
11  "valid_from": "2019-04-15T00:00:00Z",
12  "valid_until": null,
13  "evidence_anchors": [
14    "evidence://public-record/ministry-roster-2019",
15    "evidence://financial-trace/audit-report-2022",
16    "evidence://insider-source/case-127"
17  ],
18  "confidence": {
19    "source_reliability": "B", "information_validity": 2
20  },
21  "prov": {
22    "wasAttributedTo": "agent--analyst-007",
23    "wasDerivedFrom": [
24      "src--ministry-roster-001",
25      "src--audit-report-042"
26    ]
27  }
28 }
```

Listing 7: DSIX Relationship object (worked example)

B Synthetic Dataset for Case Study

The dataset that backs Section 8 is fully synthetic. We provide here the adjacency summary sufficient to reproduce the Monte Carlo results.

B.1 Nodes

Factions: F_C, F_I, F_R (3 nodes).

Actors: $|A| = 19$. Distribution by faction:

- F_C : a_{C1} (Controller), a_{C2}, a_{C3}, a_{C4} (Financiers), a_{C5}, a_{C6} (Brokers), a_{C7} (Shield, retired judge), a_{C8} (Obfuscator, media owner). The Shield a_{C7} also has **MEMBER_OF** to F_R (the contested-judge ambiguity).
- F_I : a_{I1} (Controller), a_{I2}, a_{I3} (Mobilizers), a_{I4} (Financier), a_{I5} (Agitator).
- F_R : a_{R1} (Controller, cleric), a_{R2}, a_{R3} (Hubs, regional coordinators), a_{R4}, a_{R5} (Mobilizers), a_{R6} (Agitator).

Operators: O_1 (Finance, $s = 1.5$), O_2 (Security, $s = 2.0$), O_3 (Broadcasting, $s = 1.5$), O_4 (Customs, $s = 1.0$), O_5 (Judicial, $s = 2.0$).

Fronts, Operations, Resources: 4 Fronts (F_C controls 2 commercial Fronts; F_R controls 1 charitable Front; F_I holds 1 small advocacy Front); 3 Operations (one per faction); 4 Resources; 2 Cognitive Tactics.

Total node count: $|V| = 3 + 19 + 5 + 4 + 3 + 4 + 2 = 40$.

B.2 Edges (Adjacency Summary)

MEMBER_OF (Actor $\rightarrow$ Faction): 19 edges, plus 1 dual-membership (a_{C7} also **MEMBER_OF** F_R). Total: 20.

Faction-to-Faction: 3 edges. F_I **HOSTILE_TO** F_C ; F_I **HOSTILE_TO** F_R ; F_C **LIAISES_VIA** F_R (via a_{C7}).

Faction-to-Operator (capture edges):

- F_C **INFILTRATES** O_1
- F_C **CO_OPTS** O_4
- F_C **CO_OPTS** O_5 (shared with F_R)
- F_I **CO_OPTS** O_3
- F_R **INFILTRATES** O_2
- F_R **CO_OPTS** O_5 (shared with F_C)

Within-faction co-membership edges (for resilience graph): Within each faction, Actors are connected by an undirected co-membership edge to all other Actors of the same faction. This yields:

- F_C co-membership: $\binom{8}{2} = 28$ edges
- F_I co-membership: $\binom{5}{2} = 10$ edges
- F_R co-membership: $\binom{6}{2} = 15$ edges

Cross-faction edges via a_{C7} : The dual-membership Actor a_{C7} adds bridging edges between F_C Actors and F_R Actors. For resilience analysis these are weighted at 0.5.

Faction-Front, Faction-Operation, Faction-Resource: $4 + 3 + 4 = 11$ edges.

Total edge count: $20 + 3 + 6 + 53 + 11 = 93$ edges (approximately; the precise count depends on weighting decisions for the contested edges).

B.3 Confidence Assignments

All capture edges in the case study are rated B2 ($w_c = 0.7225$) except the contested O_5 edges which are rated C3 ($w_c = 0.7 \times 0.65 = 0.455$). Within-faction co-membership edges are rated B1 ($w_c = 0.85$) for F_C (well-documented commercial relationships), C2 for F_I ($w_c = 0.595$), and B2 for F_R ($w_c = 0.7225$).

B.4 Reproducibility

The Monte Carlo random seed is $\rho = 42$. Implementations should use a deterministic pseudorandom number generator (e.g., NumPy’s PCG64 or Python’s Mersenne Twister) seeded at 42. The simulation pseudocode is given in Appendix C; a reference Python implementation is available with the supplementary material of this paper.

C Simulation Pseudocode

The following pseudocode (Python-flavored) implements Algorithm 1 (Section 7). It is designed to be a direct translation into NumPy/NetworkX. Coefficient values and seed are taken from the case study (Section 8).

```
1 import numpy as np
2 import networkx as nx
3
4 def dsix_resilience(G, M=10000, k=3,
5                     strategy="uniform",
6                     role_of=None, seed=42):
7     """Monte Carlo node-removal resilience test.
8
9     G:      networkx graph (edges may carry w_c
10            attribute for confidence weighting)
11     M:      number of iterations
12     k:      nodes removed per iteration
13     strategy: 'uniform' | 'betweenness' | 'role'
14     role_of: dict node->role (required for 'role')
15     seed:    RNG seed
16     """
17     rng = np.random.default_rng(seed)
18     V = list(G.nodes)
19     N = len(V)
20
21     # Precompute weights for sampling strategies
22     if strategy == "uniform":
23         weights = np.ones(N) / N
24     elif strategy == "betweenness":
25         bc = nx.betweenness_centrality(G,
26                                       weight="w_c", normalized=True)
27         bc_arr = np.array([bc[v] for v in V])
28         weights = bc_arr / bc_arr.sum()
29     elif strategy == "role":
30         target_role = "BROKER" # or HUB, MOBILIZER...
31         eligible = [i for i, v in enumerate(V)
32                    if role_of.get(v) == target_role]
33         weights = np.zeros(N); weights[eligible] = 1
34         weights = weights / weights.sum()
35     else:
36         raise ValueError(strategy)
37
38     lcc_ratios = np.empty(M)
39     n_components = np.empty(M, dtype=int)
40     mean_paths = np.empty(M)
41     continuity = np.empty(M, dtype=int)
42
43     for m in range(M):
44         # Sample removal set without replacement
45         idx = rng.choice(N, size=k,
46                          replace=False, p=weights)
47         R = {V[i] for i in idx}
48         Gm = G.subgraph(set(V) - R).copy()
49
50         # LCC ratio
51         if Gm.number_of_nodes() == 0:
52             lcc_ratios[m] = 0.0
53             n_components[m] = 0
54             mean_paths[m] = 0.0
55             continuity[m] = 0
56             continue
57         comps = list(nx.connected_components(Gm))
58         lcc = max(comps, key=len)
59         lcc_ratios[m] = len(lcc) / N
60         n_components[m] = len(comps)
61
62         # Mean path within LCC
63         Glcc = Gm.subgraph(lcc).copy()
64         if Glcc.number_of_nodes() >= 2:
65             mean_paths[m] = nx.average_shortest_path_length(
66                 Glcc, weight=None)
67         else:
68             mean_paths[m] = 0.0
69
70     # Command continuity: do designated source-target
```

```

71     # pairs (e.g., Controller -> top Broker) remain
72     # connected?
73     continuity[m] = check_designated_pairs(Glcc)
74
75     # Bootstrap 95% CI on the mean of each metric
76     def boot_ci(arr, B=2000):
77         boots = rng.choice(arr, size=(B,len(arr)),
78                             replace=True).mean(axis=1)
79         return np.percentile(boots, [2.5, 97.5])
80
81     return {
82         "lcc": (lcc_ratios.mean(),
83                *boot_ci(lcc_ratios)),
84         "components": (n_components.mean(),
85                         *boot_ci(n_components)),
86         "mean_path": (mean_paths.mean(),
87                       *boot_ci(mean_paths)),
88         "continuity": (continuity.mean(),
89                        *boot_ci(continuity))
90     }

```

Listing 8: DSIX Monte Carlo Resilience Test (pseudocode)

```

1  def confidence_weight(reliability, validity):
2      """FM 2-22.3 ordinal -> [0,1] product weight."""
3      wr = {"A":1.00,"B":0.85,"C":0.70,
4            "D":0.50,"E":0.30,"F":0.10}[reliability]
5      ww = {1:1.00, 2:0.85, 3:0.65,
6            4:0.40, 5:0.20, 6:0.10}[validity]
7      return wr * ww
8
9  def build_dsix_graph(sdos, sros):
10     """Convert DSIX bundle to NetworkX graph with
11     edges weighted by confidence."""
12     G = nx.Graph()
13     for obj in sdos:
14         G.add_node(obj["id"], **obj)
15     for rel in sros:
16         c = rel.get("confidence", {})
17         wc = confidence_weight(
18             c.get("source_reliability","C"),
19             c.get("information_validity",3))
20         G.add_edge(rel["source_ref"],
21                   rel["target_ref"],
22                   w_c=wc,
23                   rel_type=rel["relationship_type"])
24     return G

```

Listing 9: Confidence-weighted edge construction

The full reference implementation, including the case-study graph construction and the OCR/RA/CD/SR computation routines, is contemplated as supplementary material accompanying any published version of this paper.

C.1 Neo4j Property-Graph Import

For deployments using a property-graph backend, the following Cypher fragment maps a DSIX JSON bundle to Neo4j nodes and relationships. The pattern preserves `w_c`, validity intervals, and provenance as edge properties, permitting confidence-weighted queries on the resulting graph.

```

1  // Step 1: import SDOs as nodes (one CREATE per SDO type)
2  UNWIND $sdos AS o
3  CALL apoc.create.node(
4      [o.type],
5      o {.*, _id: o.id}
6  ) YIELD node
7  RETURN count(node);
8
9  // Step 2: import SROs as typed relationships
10 UNWIND $sros AS r
11 MATCH (s {_id: r.source_ref})
12 MATCH (t {_id: r.target_ref})
13 CALL apoc.create.relationship(
14     s, r.relationship_type, {
15         capture_intensity: r.capture_intensity,
16         valid_from: r.valid_from,
17         valid_until: r.valid_until,
18         w_c: r.confidence.w_c,
19         source_reliability: r.confidence.source_reliability,
20         information_validity: r.confidence.information_validity,
21         affiliation_root: r.affiliation_root,

```

```

22     evidence_anchors: r.evidence_anchors,
23     prov: r.prov
24 }, t
25 ) YIELD rel
26 RETURN count(rel);
27
28 // Step 3: example confidence-weighted OCR query
29 MATCH (f:faction {name:"F_C"})-[r]->(o:operator)
30 WHERE r.type IN ['CO_OPTS','INFILTRATES','CONTROLS']
31 WITH o, r,
32     CASE r.type
33         WHEN 'CO_OPTS' THEN 1.0/3
34         WHEN 'INFILTRATES' THEN 2.0/3
35         WHEN 'CONTROLS' THEN 1.0
36     END AS intensity
37 RETURN sum(intensity * r.w_c * o.strategic_weight)
38 / sum(o.strategic_weight) AS OCR;

```

Listing 10: Neo4j Cypher import skeleton for a DSIX bundle

Schema validation against the DSIX JSON Schema sketch (Appendix A) can be executed with any Draft 2020-12-compliant validator. A representative invocation using the Python `jsonschema` library is:

```

1 import json, jsonschema
2 from jsonschema import Draft202012Validator
3
4 with open("dsix_v1_0.json") as fs, open("bundle.json") as fb:
5     schema = json.load(fs)
6     bundle = json.load(fb)
7
8 validator = Draft202012Validator(schema)
9 errors = sorted(validator.iter_errors(bundle),
10                key=lambda e: e.path)
11 for e in errors:
12     print(f"{list(e.path)}: {e.message}")

```

Listing 11: JSON Schema validation invocation

D 1MDB Worked Encoding: Full JSON Bundle

The following bundle is the full JSON form of the 1MDB worked encoding summarized in Tables 9 and 10 (Section 9-B). It is offered as a concrete proof-of-concept of how a real-world case is rendered in DSIX. All civil-forfeiture-based assertions carry `legal_status: asserted_in_complaint`; the operator-control assertion grounded in the Federal Court of Malaysia 2022 SRC ruling carries `legal_status: judicially_confirmed`. Identifiers use placeholder UUIDs.

```

1 {
2   "spec_version": "1.0",
3   "bundle_id": "bundle--1mdb-pof-001",
4   "sdos": [
5     { "type": "faction",
6       "id": "faction--1mdb-najib-jho",
7       "name": "F_1MDB Najib Jho Low patronage",
8       "founding_year": 2009,
9       "primary_affiliation_roots": [
10        "PATRON_CLIENT", "STRATEGIC_MARRIAGE", "CARTEL"],
11       "current_phase": "DEEP_STATE",
12       "confidence": {"source_reliability": "B",
13                     "information_validity": 2}
14     },
15     { "type": "actor",
16       "id": "actor--najib-razak",
17       "name": "Najib Razak",
18       "biographical": {"birth_year": 1953,
19                       "nationality": "MY"},
20       "confidence": {"source_reliability": "A",
21                     "information_validity": 1}
22     },
23     { "type": "actor",
24       "id": "actor--jho-low",
25       "name": "Low Taek Jho",
26       "aliases": ["Jho Low"],
27       "biographical": {"birth_year": 1981,
28                       "nationality": "MY"},
29       "confidence": {"source_reliability": "B",
30                     "information_validity": 2}
31     },
32     { "type": "operator",

```



```

33     "id": "operator--1mdb",
34     "name": "1Malaysia Development Berhad",
35     "operator_class": "STATE_ENTERPRISE",
36     "strategic_weight": 1.5,
37     "confidence":{"source_reliability":"A",
38                   "information_validity":1}
39 },
40 {
41   "type": "operator",
42   "id": "operator--mof-my",
43   "name": "Malaysian Ministry of Finance",
44   "operator_class": "FINANCE_MINISTRY",
45   "strategic_weight": 1.5,
46   "confidence":{"source_reliability":"A",
47                 "information_validity":1}
48 },
49 {
50   "type": "front",
51   "id": "front--good-star",
52   "name": "Good Star Limited",
53   "jurisdiction":"SC",
54   "registry_id":"GS-2009-XXX",
55   "confidence":{"source_reliability":"B",
56                 "information_validity":2}
57 },
58 {
59   "type": "front",
60   "id": "front--tanore",
61   "name": "Tanore Finance Corporation",
62   "jurisdiction":"VG",
63   "confidence":{"source_reliability":"B",
64                 "information_validity":2}
65 },
66 {
67   "type": "resource",
68   "id": "resource--petrosaudi-proceeds-2009",
69   "name": "PetroSaudi JV diverted proceeds",
70   "valuation_usd": 1830000000,
71   "valuation_basis":"DOJ_complaint",
72   "confidence":{"source_reliability":"B",
73                 "information_validity":2}
74 },
75 {
76   "type": "operation",
77   "id": "operation--petrosaudi-diversion",
78   "name": "PetroSaudi JV diversion phase",
79   "start_date":"2009-09-01",
80   "end_date": "2009-10-31",
81   "confidence":{"source_reliability":"B",
82                 "information_validity":2}
83 },
84 {
85   "type": "cognitive_tactic",
86   "id": "tactic--sarawak-block-2015",
87   "name": "Sarawak Report blocking and fake-news framing",
88   "tactic_family":"NARRATIVE_SUPPRESSION",
89   "start_date":"2015-07-01",
90   "confidence":{"source_reliability":"C",
91                 "information_validity":3}
92 }
93 ]
94 }

```

Listing 12: 1MDB DSIX bundle: SDO array (objects 1 through 10)

```

1 {
2   "sros": [
3     {
4       "id": "sro--01", "type": "dsix-relationship",
5       "relationship_type": "MEMBER_OF",
6       "source_ref": "actor--najib-razak",
7       "target_ref": "faction--1mdb-najib-jho",
8       "valid_from": "2009-07-01",
9       "legal_status": "judicially_confirmed",
10      "evidence_anchors": ["court_record:FC-MY-2022-SRC"],
11      "confidence":{"source_reliability":"B",
12                    "information_validity":2}
13    },
14    {
15      "id": "sro--02", "type": "dsix-relationship",
16      "relationship_type": "MEMBER_OF",
17      "source_ref": "actor--jho-low",
18      "target_ref": "faction--1mdb-najib-jho",
19      "valid_from": "2009-07-01",
20      "legal_status": "asserted_in_complaint",
21      "evidence_anchors": ["court_record:DOJ-1MDB-2017",
22                           "public_record:PAC-MY-2016"],
23      "confidence":{"source_reliability":"B",
24                    "information_validity":2}
25    },
26    {
27      "id": "sro--03", "type": "dsix-relationship",
28      "relationship_type": "HOLDS_ROLE_IN",

```

```

26     "source_ref": "actor--najib-razak",
27     "target_ref": "operator--mof-my",
28     "valid_from": "2009-04-09",
29     "valid_until": "2018-05-10",
30     "role_title": "Prime Minister and Finance Minister",
31     "legal_status": "public_record",
32     "evidence_anchors": ["public_record:official-gazette"],
33     "confidence": {"source_reliability": "A",
34                   "information_validity": 1}
35 },
36 { "id": "sro--04", "type": "dsix-relationship",
37   "relationship_type": "HOLDS_ROLE_IN",
38   "source_ref": "actor--najib-razak",
39   "target_ref": "operator--1mdb",
40   "valid_from": "2009-07-01",
41   "role_title": "Chair, Board of Advisors",
42   "legal_status": "corporate_registry",
43   "evidence_anchors": ["public_record:1MDB-charter"],
44   "confidence": {"source_reliability": "A",
45                 "information_validity": 1}
46 },
47 { "id": "sro--05", "type": "dsix-relationship",
48   "relationship_type": "CO_OPTS",
49   "source_ref": "faction--1mdb-najib-jho",
50   "target_ref": "operator--mof-my",
51   "capture_intensity": 0.333,
52   "valid_from": "2009-07-01",
53   "legal_status": "asserted_in_complaint",
54   "evidence_anchors": ["court_record:DOJ-1MDB-2017"],
55   "confidence": {"source_reliability": "B",
56                 "information_validity": 2}
57 },
58 { "id": "sro--06", "type": "dsix-relationship",
59   "relationship_type": "CONTROLS",
60   "source_ref": "faction--1mdb-najib-jho",
61   "target_ref": "operator--1mdb",
62   "capture_intensity": 1.0,
63   "valid_from": "2009-07-01",
64   "legal_status": "judicially_confirmed",
65   "evidence_anchors": ["court_record:FC-MY-2022-SRC",
66                       "court_record:DOJ-1MDB-2017"],
67   "confidence": {"source_reliability": "B",
68                 "information_validity": 2}
69 },
70 { "id": "sro--07", "type": "dsix-relationship",
71   "relationship_type": "USES",
72   "source_ref": "faction--1mdb-najib-jho",
73   "target_ref": "front--good-star",
74   "valid_from": "2009-09-01",
75   "legal_status": "asserted_in_complaint",
76   "evidence_anchors": ["court_record:DOJ-1MDB-2017"],
77   "confidence": {"source_reliability": "B",
78                 "information_validity": 2}
79 },
80 { "id": "sro--08", "type": "dsix-relationship",
81   "relationship_type": "USES",
82   "source_ref": "faction--1mdb-najib-jho",
83   "target_ref": "front--tanore",
84   "valid_from": "2013-03-01",
85   "legal_status": "asserted_in_complaint",
86   "evidence_anchors": ["court_record:DOJ-1MDB-2017"],
87   "confidence": {"source_reliability": "B",
88                 "information_validity": 2}
89 },
90 { "id": "sro--09", "type": "dsix-relationship",
91   "relationship_type": "HOLDS",
92   "source_ref": "front--good-star",
93   "target_ref": "resource--petrosaudi-proceeds-2009",
94   "valid_from": "2009-09-30",
95   "legal_status": "asserted_in_complaint",
96   "evidence_anchors": ["financial_trace:DOJ-1MDB-2017"],
97   "confidence": {"source_reliability": "B",
98                 "information_validity": 2}
99 },
100 { "id": "sro--10", "type": "dsix-relationship",
101   "relationship_type": "EMPLOYS",
102   "source_ref": "operation--petrosaudi-diversion",
103   "target_ref": "tactic--sarawak-block-2015",
104   "valid_from": "2015-07-19",
105   "legal_status": "observational",
106   "evidence_anchors": ["public_record:MCMC-block-order"],
107   "confidence": {"source_reliability": "C",
108                 "information_validity": 3}
109 }
110 ]

```

Listing 13: 1MDB DSIX bundle: SRO array (relationships 1 through 10)

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